Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

Anonymous Author(s)
Affiliation
Address
email

Abstract

Large language models are increasingly deployed in decisions that turn on moral judgement, from content moderation to clinical prioritisation, yet they answer as if the whole world thinks like the West. The Moral Machine experiment [Awad et al., 2018] showed this is wrong at scale: 40 million judgments across 233 countries reveal that moral preferences are systematically structured by culture, and a model that ignores this variation does not merely underperform. It imposes one society's intuitions on all others. Existing fixes do not scale to global deployment: fine-tuning needs per-country preference data and GPU budgets, reward-guided decoding needs per-country reward models, and activation steering needs access to model internals that black-box APIs do not expose. We work in the only regime that scales: inference time, with no weight updates, no training data, and no internal access. The key observation is that within-country demographic *disagreement*, not consensus, is the steering signal: when culturally grounded personas agree, the base model is already calibrated; when they disagree, the spread tells us what to fix and by how much. DISCA (Disagreement-Informed Steering for Cultural Alignment) instantiates each country as a panel of four World-Values-Survey-grounded persona agents [Haerpfer et al., 2022], converts their disagreement into a bounded, loss-averse correction whose magnitude is set by the panel's variance, and shrinks the correction toward zero when the estimate is unreliable. Across 20 countries and 7 open-weight backbones (2B–70B) from five model families, DISCA reduces cultural misalignment on MultiTP [Jin et al., 2025] by 10–24% on binary moral dilemmas and 2–7% on open-ended scenarios; a 14B backbone with DISCA reaches lower absolute misalignment than a vanilla 70B model.

1 Introduction

Large language models increasingly mediate decisions that turn on moral judgement: content moderation policies, clinical prioritisation guidelines, autonomous-vehicle behaviour specifications, and high-stakes recommendation pipelines. In each of these settings, the model's implicit moral preferences become the system's moral preferences. Yet a growing body of evidence shows that these implicit preferences are not culturally neutral. The Moral Machine experiment collected 40 million judgments across 233 countries and showed that moral preferences are systematically structured by culture [Awad et al., 2018]; LLM outputs correlate most strongly with respondents from Western, educated, industrialised, rich, and democratic populations [Henrich et al., 2010, Santurkar et al., 2023]; and post-training alignment introduces further unintended cultural shifts [Ryan et al., 2024, Zewail et al., 2026]. A model that ignores this variation does not merely underperform on a benchmark; it imposes one society's intuitions on every user it serves.

The cultural gap is now well-documented at the model level. The MultiTP benchmark [Jin et al., 2025] extends the Moral Machine protocol to 107 languages with country-specific Average Marginal

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Component Effects (AMCEs), and reports that none of 19 evaluated LLMs recovers the country-level structure of human moral preferences. The asymmetries are substantial: Japan ranks first among countries on preference for sparing pedestrians, China ranks 116th, and current models do not capture either extreme. Concurrent work shows that sociodemographic persona prompts shift LLM moral decisions two to four times more than they shift human decisions [**?**], indicating that the models are not insensitive to cultural framing, but that the framing they receive at deployment time does not match the cultural distribution they are asked to serve.

Closing this gap is harder than it looks. Three families of methods have been proposed, and each demands resources that do not scale to the long tail of countries served by a single deployed model. Fine-tuning and culture-aware adapters require per-country preference datasets and GPU budgets for every target population [Zhang et al., 2026, Yao et al., 2025]; reward-guided decoding requires a separate trained reward model per country [Khanov et al., 2024, Mudgal et al., 2024]; and activation steering requires write access to model internals [Arditi et al., 2025, Zou et al., 2023, Turner et al., 2024], which black-box APIs do not expose. Any deployable solution must therefore work at inference time, must not assume per-country reward models or labelled preference data, and must not assume access to model internals beyond what an API exposes. We refer to this as the black-box, public-data-only regime, and argue that it is the only regime in which cultural alignment can scale from a handful of curated countries to the hundreds of cultures a globally-deployed model actually serves.

We make the following observation. When certain culturally grounded persona prompts representing the same country face the same moral dilemma, the dispersion of their decoding-time logit gaps is itself informative. If they agree, the base model is already calibrated for that country and no correction is needed; if they disagree, the spread of their gaps encodes both the direction and the magnitude of the needed correction. Our method, DISCA (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022], evaluates them in a single batched forward pass, and converts the panel's variance into a bounded, loss-averse logit correction whose magnitude is automatically attenuated when the underlying estimate is unreliable. We show that this attenuation is not heuristic: it implements a variance-aware shrinkage estimator whose shrinkage weight is determined by the panel's empirical disagreement, formally characterising when the correction can be trusted.

Across 20 countries spanning four continents and seven open-weight backbones (2B–70B parameters) from five model families, DISCA reduces cultural misalignment on MultiTP by 10–24% on binary moral dilemmas and 2–7% on open-ended scenarios. The strongest absolute alignment is achieved by Phi-4 (14B), which surpasses a vanilla Llama-3.3-70B backbone despite using one fifth of the parameters; this suggests that, in the regime we study, calibration competes with scale rather than merely complementing it. An evaluation on the BLEnD factual cultural QA benchmark [Myung et al., 2024] reveals a clean scope boundary: DISCA steers values, where the decision reduces to a scalar logit gap, but does not transfer to factual recall, where the decision is a single token in a large vocabulary. Failure modes are architecturally rather than culturally determined and are diagnosable from the base model's vanilla logit conditioning alone.

Contributions.

- An inference-time cultural alignment method that requires no weight updates, no per-country reward models, and no white-box access.

- A formal characterisation of the disagreement-driven correction as a variance-aware shrinkage estimator, where panel variance determines the shrinkage weight and a dual-pass reliability gate provides a finite-sample plug-in estimator of this weight.

- An empirical evaluation across 20 countries, four continents, seven open-weight backbones, and three evaluation formats (binary moral dilemmas, open-ended scenarios, factual QA), demonstrating that a 14B model with the proposed recipe matches or surpasses a 70B model without it.

2 Related Work

Benchmarks and cultural evaluation. The Moral Machine experiment [Awad et al., 2018] established that moral preferences are culturally structured; MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs. Takemoto [2024] and **?** showed that LLM alignment varies widely across model families. **?** demonstrated that sociodemographic personas induce moral-decision shifts in LLMs 2–4$\times$ larger than in humans, with political orientation producing the largest effect (a “partisan sorting” phenomenon absent in human respondents). Their finding that simple persona prompts amplify rather than correct misalignment directly motivates DISCA's approach: we use WVS-grounded personas not to steer decisions toward a single viewpoint but to measure within-country disagreement and convert it into a bounded correction. Existing moral benchmarks remain English-centric [Hendrycks et al., 2021, Jin et al., 2022, Nie et al., 2023] or cover fewer languages [Kirk et al., 2024, Durmus et al., 2023, Qiu et al., 2025]. We validate on MultiTP's binary trolley vignettes, the BLEnD factual cultural QA benchmark [Myung et al., 2024] (Appendix A6), and a dedicated open-ended moral-dilemma track (§5). On the diagnostic side, LLM outputs correlate most with WEIRD respondents [Santurkar et al., 2023], RLHF introduces unintended cultural shifts [Ryan et al., 2024], and recent work shows LLMs systematically stereotype non-Western moral values [Zewail et al., 2026].

Inference-time alignment. The pluralistic alignment vision [Sorensen et al., 2024] argues that inference-time methods are a natural vehicle for cultural adaptation, but existing approaches each sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs per-country reward models; culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] need per-culture training. Recent logit-level steering [An et al., 2026, Khanuja et al., 2026] demonstrates that the logit surface is a richer control interface than appreciated, but neither targets country-conditioned moral calibration. Black-box alignment modules [Bobbili et al., 2025, Chen et al., 2025] still require a training stage, and adaptive IS in pre-logit space [Kanai et al., 2025] requires activation access. Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] applies the same PT value shape we use, but at training time; DISCA applies it at *test time*, per scenario, aggregating heterogeneous personas rather than shaping a trained policy.

Distinction from superficially similar paradigms. DISCA may superficially resemble ensemble methods, but the resemblance is architectural, not algorithmic. In self-consistency [**?**] and LLM debate [**??**], diversity is noise to be eliminated via majority vote; in standard ensembles, multiple outputs are averaged to reduce variance of the *prediction*. In DISCA, diversity *is* the signal: the *spread* of persona logit gaps, not the mode, determines the correction magnitude. This distinction is not merely conceptual. We prove (Theorem 1) that the disagreement-driven correction is a James–Stein shrinkage estimator [**?**] that strictly dominates unshrunk consensus in worst-case MSE: ensembles average to reduce output variance, whereas DISCA uses within-panel variance as a sufficient statistic to *control* correction magnitude. Standard calibration (temperature scaling, MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]) adjusts confidence isotropically; DISCA applies anisotropic, loss-averse corrections. The ablation (Table 5) confirms the distinction empirically: removing personas degrades MIS by +0.047–0.061, while removing the reliability gate loses +0.015–0.021, effects no ensemble or calibration method reproduces.

To our knowledge, DISCA is the first training-free method targeting country-level moral calibration. The method is grounded in control-as-inference, Prospect Theory, and importance-sampling variance control (Appendix A12).

3 Method: DISCA

DISCA produces a small adjustment to the model's decision-token logits, computed from a number of culturally grounded personas and applied at decode time. The size of the adjustment is not fixed: when these personas agree on what the country's preference should be, we apply the adjustment in full; when they disagree, we apply only a fraction of it, scaled by how much the panel disagrees. The rest of this section makes this idea precise. We first fix notation in 3.1, then formalise the principle that within-panel disagreement controls the right size of the correction in **??**, specify the three components that compute this correction in practice **??**, and finally discuss what the formalisation delivers and what it does not **??**.

3.1 Setup and notation

We work on the MultiTP benchmark [Jin et al., 2025], which adapts the Moral Machine trolley-problem protocol to language-model evaluation. Each scenario x describes two groups of people the autonomous vehicle could spare and asks the model to choose between them via decision tokens A and B. Every scenario isolates one of six moral attributes: species (humans vs. animals), gender, age, physical fitness, social value, and utilitarianism (more vs. fewer lives saved). The two groups in a scenario differ on the isolated attribute and are matched on the other five. Each scenario therefore *belongs to* the attribute it isolates, partitioning the scenario pool into six attribute groups.

For target country c, let $\mathcal{S}_d^{(c)}$ denote the country-c scenarios that isolate attribute d. Across many scenarios, the country's preference on attribute d can be summarised by averaging the rate at which respondents choose to spare the preferred group; this average is called the Average Marginal Component Effect (AMCE) on attribute d. Stacking the six AMCEs gives a six-dimensional preference vector. Let $h^{(c)} \in \mathbb{R}^6$ denote the human AMCE vector for country c, aggregated from 40M+ Moral Machine judgments [Awad et al., 2018, Jin et al., 2025].

A frozen LLM f_θ produces logits $z(x) = [z_a, z_b]$ at the decision-token position. We define the decision gap $\delta = z_b - z_a$ and the sparing probability $p_{\text{spare}} = \sigma(\delta / T_{\text{dec}})$, where T_{dec} is a decoding temperature. The model's AMCE on attribute d is the average sparing probability over scenarios in $\mathcal{S}_d^{(c)}$:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{x \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(x). \tag{1}$$

Stacking across the six attributes gives the model's AMCE vector $\hat{m}^{(c)} \in \mathbb{R}^6$, which we compare against $h^{(c)}$.

Our goal is to compute, from f_θ alone, a correction $\delta^\star$ to add to the base-model gap δ_{base} such that the corrected AMCE vector $\hat{m}^{(c)}(\theta, \delta^\star)$ is closer to $h^{(c)}$ in ℓ_2 distance. We refer to this distance as the misalignment score (MIS), the primary metric of the paper. The next subsection states the principle that governs how large $\delta^\star$ should be.

3.2 Cultural Persona Construction

We use $N{=}4$ personas (three age cohorts plus a country-wide aggregate) grounded in WVS Wave 7 microdata [Haerpfer et al., 2022]. Four is the smallest panel covering the three demographic axes with most within-country variance (young, middle-aged, older) while preserving a population anchor [Inglehart and Welzel, 2005, Kish, 1965]. Ablation confirms the panel is load-bearing: removing it costs $+0.083$ MIS, the largest single-component drop (Table 5). A sensitivity check replacing the aggregate persona with a utilitarian anchor yields marginal $\Delta{=}{-}0.02$ MIS (Appendix A15.4).

For each country c, following the 10-variable cultural-value scheme of Greco et al. [2026], we extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the "inverted" WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts-*young* (<36), *middle-aged* (36–55), and *older* (>55)-yielding three cohort-specific personas that capture generational variation in cultural values. A fourth persona aggregates all respondents regardless of age, anchoring the ensemble with the country's population-wide cultural centre of gravity. All four personas are thus data-driven and country-specific; no agent carries a hard-coded moral stance that could bias particular dimensions. Dimension means are normalised to $[0, 1]$ and mapped to four-level natural-language descriptors via quartile cuts (≥ 0.75, ≥ 0.50, ≥ 0.25, < 0.25); see Appendix A15. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. The 10-dimensional WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance across the 20-country panel (Appendix A15.3), and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones. Crucially, the IS stage automatically down-weights persona signals that increase collective disutility, acting as a built-in noise filter (Appendix A15.3). A WVS dropout study confirms a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing (ΔMIS > 0.03), while national pride and life satisfaction have near-zero impact (Appendix A7.4). Persona examples are in Appendix A15.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix A10.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here δ_{base} is the base-prompt decision gap; r_i measures how strongly persona i shifts the model away from the neutral baseline. (The scalar r in Eq. 6 denotes the bootstrap reliability weight, not a reward.)

3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A $\leftrightarrow$ B) and group labels (Group A $\leftrightarrow$ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields δ_{base} under the same rule. Implementation details (placeholder-based three-step swap to avoid double-substitution bugs in native scripts) are in Appendix A14.5.

3.5 From Persona Disagreement to a Bounded Correction

The persona consensus $\bar{\delta}$ already captures the cultural direction, but applying it directly would ignore two problems: the consensus may overcorrect toward one loud persona, and finite-sample noise may push the correction in an unreliable direction. We address both through importance sampling with a cooperative, loss-averse utility.

We sample candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ around the consensus and score each by how much it improves alignment with every persona, not just the loudest one. Per-persona gains are aggregated through a Prospect-Theory value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] with $\kappa{\approx}2.25$, encoding a "do no harm" prior: the PT kernel is a *controller design choice*, an asymmetric aggregation kernel where harming one demographic group is penalised more than equally-sized help to another, not a literal model of LLM cognition. Auxiliary runs with linear and WVS-weighted kernels underperform the PT kernel, and robustness sweeps keep all JSD spans within ± 0.004 (Appendix A13). The softmax-weighted IS update can be read as approximate control-as-inference [Williams et al., 2018, Levine, 2018].

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}})\,\frac{1}{N}\sum_{i=1}^{N} v\!\left(\frac{g_{i,k}}{\sigma}\right) \;+\; \lambda_{\text{coop}}\, v\!\left(\frac{g_{\text{cons},k}}{\sigma}\right), \tag{5}$$

where $v(\cdot)$ is the Kahneman–Tversky value function with gain curvature $\alpha{=}0.88$ and loss aversion $\kappa{=}2.25$ [Kahneman and Tversky, 1979], $g_{i,k}$ measures how much candidate k improves over the base for persona i (Appendix A14), and $\lambda_{\text{coop}}{=}0.7$ weights the group consensus. Candidates are then softmax-weighted by utility and combined into a single correction $\delta^\star_m$; an effective-sample-size guard zeros the correction when the weights collapse, falling back on the raw consensus (Appendix A14).

Sampling noise is handled by a *dual-pass reliability gate*: DISCA splits the IS budget into two independent passes and compares them:

$$\delta^\star_{\rm IS} \;=\; \underbrace{\exp\!\big(-(\delta^\star_1-\delta^\star_2)^2/s\big)}_{r\,\in\,(0,1]:\ \text{reliability weight}} \cdot \frac{\delta^\star_1+\delta^\star_2}{2}. \tag{6}$$

When the two passes concur, $r \approx 1$ and the full correction applies; when they diverge, r decays exponentially, requiring no hard threshold or manual tuning. This self-limiting property prevents confident-but-wrong corrections on noisy logit landscapes and provides the safety margin the open-ended extension (§3.6) relies on. The dual-pass split yields a bootstrap-style variance probe [Efron, 1979, Elvira and Martino, 2021] that flags instability rather than hiding it behind more samples, with an ESS guard [Kish, 1965] zeroing the offset when weights collapse. We treat the gate as a variance heuristic, not a formal unbiasedness certificate (Appendices A13–A14). The final correction combines the reliability-gated IS offset with an interpolation between persona consensus and the debiased base (Appendix A14).

This self-regulating behaviour has a precise formal justification.

Theorem 1 (Optimality of disagreement-driven shrinkage)**.** *Let $\delta_1,\dots,\delta_N$ be the persona logit gaps for a single scenario, modelled as $\delta_i=\delta_h+\eta_i$ with η_i i.i.d., $\mathbb{E}[\eta_i]=0$, $\mathrm{Var}(\eta_i)=\tau^2$, and δ_h the unknown human-preferred gap. Define the consensus correction $\Delta=\bar\delta-\delta_{\rm base}$ and the within-panel disagreement $D^2=\frac{1}{N-1}\sum_i(\delta_i-\bar\delta)^2$. Then:*

(i) ***Disagreement is a sufficient statistic for correction risk.*** *The MSE decomposes as $\mathbb{E}[(\Delta-\Delta_h)^2]=\bar b^2+\tau^2/N$, where $\bar b$ is the mean persona bias and D^2 is an unbiased estimator of τ^2.*

(ii) ***Shrinkage dominates consensus.*** *For $N\ge 4$, the James–Stein shrinkage estimator $\hat\Delta_{\rm JS}=\max\!\big(0,\ 1-\frac{(N-3)\,D^2}{N\,\Delta^2}\big)\cdot\Delta$ strictly dominates the unshrunk consensus Δ in MSE uniformly over δ_h.*

(iii) ***The reliability gate implements asymptotically optimal shrinkage.*** *The dual-pass weight $r=\exp\!\big(-(\delta^\star_1-\delta^\star_2)^2/s\big)$ is a monotone decreasing function of the inter-replicate variance $V_r=(\delta^\star_1-\delta^\star_2)^2/2$, itself an unbiased estimator of $\tau^2/K_{\rm half}$. As $K_{\rm half}\to\infty$, $r\cdot\Delta\to\hat\Delta_{\rm JS}$ in probability.*

Proof sketch. Part (i) follows from the bias-variance decomposition and the unbiasedness of the sample variance. Part (ii) is a direct application of the James–Stein theorem [**?**]: for $N\ge 4$, shrinking a d-dimensional normal mean toward the origin dominates the MLE, and the persona model reduces to this setting with $d=N-1\ge 3$. Part (iii) follows because V_r is an unbiased estimator of $\tau^2/K_{\rm half}$ (the two half-budget passes are independent), and the exponential map $r=e^{-V_r/s}$ is a consistent plug-in for the oracle shrinkage factor $\gamma^\star=\Delta_h^2/(\Delta_h^2+\tau^2/N)$ under standard regularity (full proof in Appendix A1).

Corollary 2. *$N=4$ is the minimum panel size guaranteeing that the disagreement-driven correction dominates naive consensus in worst-case MSE. DISCA is therefore not an ensemble (which targets the mean) but a* shrinkage estimator *that uses within-group disagreement to control correction magnitude, recovering an instance of the James–Stein phenomenon for inference-time alignment.*

The final spare probability is $p_{\rm spare}=\sigma(\delta_{\rm final}/T_{\rm dec})$ with $T_{\rm dec}$=0.5. Algorithm 1 summarises the full procedure; numerical values for all symbols appear in Table 18 (Appendix A10).

3.6 Extension to Open-Ended Ethical Scenarios

To extend DISCA to open-ended scenarios, the model generates free-form responses under both the original prompt and four persona-specific prompts. An LLM then parses these outputs to return a {choice, confidence} pair. The continuous decision signal is represented by a pseudo-logit-gap (the A/B logit distance) extracted by the LLM . The dual-pass PT–IS mechanism is applied similarly to the binary case.

Algorithm 1 DISCA Inference: Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^{N}$; hyperparameters $K_{\text{half}}, \sigma, \lambda, \eta, T_{\text{dec}}, s$
Ensure: $p_{\text{spare}} \in [0, 1]$
for order $\in \{\text{orig}, \text{swap}\}$ **do**
  Batched forward: f_θ on base + N personas
  Extract logit pairs; compute $\delta_i^{(\text{ord})}$
end for
$\delta_i \leftarrow (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$
Compute $\bar{\delta}$, r_i [Eqs. 3–4]
for $m \in \{1, 2\}$ **do**
  for $k = 1$ **to** K_{half} **do**
    Sample $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$; $U_{\text{tot}}(\epsilon_k)$
  end for
  $\delta_m^\star \leftarrow \sum_k w_k^{(m)} \epsilon_k$ (if ESS $> \rho_{\text{eff}}$)
end for
$r \leftarrow \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$
$\delta_{\text{IS}}^\star \leftarrow r(\delta_1^\star + \delta_2^\star)/2$
$\delta_{\text{final}} \leftarrow \text{blend}(\bar{\delta}, \delta_{\text{base}}, \delta^\star)$
return $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$

Table 1: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_c
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 1), fixing 20 countries spanning four continents. After deduplication, quality filtering, category capping (max 80 per dimension), and oversampling (min 36 for stable AMCE estimation), each country has 310–500 scenarios (Appendix A17). A no-cap sensitivity run confirms results are robust (Appendix A11).

4.2 Models

We evaluated 28 model–method combinations spanning 12 architectures (270M–70B; Appendix A5). The main results focus on seven open-weight checkpoints with robust gains: **Llama-3.3-70B**, **Magistral-24B**, **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, **Phi-3.5-mini** (3.8B), and **Gemma-4-E2B** (2B). Models that degraded share a common pattern: already-low vanilla MIS leaving insufficient headroom (§6). All scenarios are presented in the native language; only the decision tokens (A/B) remain English (Appendix A18). The released configuration was selected from 25 variants on a three-backbone, five-country prototyping panel (Appendix A21).

4.3 Baselines and metrics

We evaluate DISCA against two tiers of baselines on the 20-country Phi-4 grid (Table 2; implementation details in Appendix A19). The first tier requires no human AMCE access: vanilla decoding (generic system prompt), WVS Profile Prompt (country-specific WVS summary), PRISM-Style Prompt [Kirk et al., 2024] (structured cultural framing), fixed logit offset, activation steering [Arditi et al., 2025, Zou et al., 2023], and MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026]. Reward-guided methods [Khanov et al., 2024, Mudgal et al., 2024] are excluded because they require

Table 2: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Oracle baselines (below dashed line) use human AMCE during calibration; even so they fail to match DISCA.

Method	AMCE?	MIS ↓
DISCA (ours)	No	**0.346**
PRISM-Style Prompt [Kirk et al., 2024]	No	0.384
MC-Dropout [Gal and Ghahramani, 2016]	No	0.403
Activation steering [Arditi et al., 2025]	No	0.430
Fixed logit offset	No	0.439
WVS Profile Prompt	No	0.453
Vanilla decoding	No	0.454
Margin scaling [Guo et al., 2017]	Yes	0.506
Temp. scaling [Guo et al., 2017]	Yes	0.513

Table 3: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean ℓ_2 AMCE distance ($\pm$ std over 3 seeds); "Win" = countries with lower DPBR MIS than vanilla.

Model	MIS ↓	Gain (%)	Win/20	r
Llama-3.3-70B	$.668_{\pm .006}$	$+21.3_{\pm 0.8}$	**20**	-0.01
Magistral-24B	$.350_{\pm .005}$	$+13.0_{\pm 0.7}$	16	+0.62
Phi-4 (14B)	$\mathbf{.346}_{\pm .004}$	$\mathbf{+23.6}_{\pm 0.6}$	18	+0.56
Qwen3-VL-8B	$.466_{\pm .006}$	$+18.8_{\pm 0.9}$	**20**	+0.22
Qwen2.5-7B	$.362_{\pm .005}$	$+20.0_{\pm 0.8}$	17	+0.45
Phi-3.5-mini	$.571_{\pm .006}$	$+10.3_{\pm 1.0}$	17	-0.35
Gemma-4-E2B (2B)	.479	+3.4	14	+0.46

per-country reward models that do not exist. The second tier is oracle baselines with human AMCE access: per-country temperature and margin scaling [Guo et al., 2017], which fit a scalar directly on the country's human AMCE and represent an upper bound on population-level scalar fits.

The primary metric is MIS (misalignment score): the ℓ_2 distance between the model's six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], macro-averaged over the 20 countries with equal weight. Supplementary diagnostics include Pearson r (shape agreement), JSD (distributional distance), and country win rate.

5 Results

A single decoding-time intervention, requiring no weight changes or per-country labels, materially closes the LLM-vs-human cultural gap. Across seven open checkpoints spanning five model families, DISCA cuts macro MIS by 10–24% relative to vanilla (Table 3), with the two strongest backbones winning on all 20 countries. PCA projection of the six-dimensional AMCE vectors (Figure 2, Appendix A4) confirms the pattern visually: every country point migrates from the vanilla cloud toward the human cloud. No simpler intervention comes close. On the same Phi-4 grid (Table 2), DISCA outperforms every training-free baseline by a wide margin, and even oracle baselines that fit a scalar directly on the human AMCE overfit rather than recover cultural structure.

The per-country breakdown (Table 7, Appendix A2) shows that the macro gain is not driven by a handful of lucky cells. Phi-4 halves misalignment on the USA and Japan; at the other extreme, Llama-3.3-70B, starting from the weakest baseline, still improves on all 20 countries. Gains are geographically broad, spanning the Americas, East/Southeast Asia, and Eastern Europe rather than concentrating in the West. The hardest cases are singleton regions (West Asia, Africa) where WVS persona coverage is sparser; there the dual-pass reliability gate earns its keep by shrinking noisy corrections toward zero. Robustness checks across nine independent axes and three random seeds confirm that all conclusions are stable well within the bootstrap noise floor (Appendices A10, A7.1).

Perhaps the most revealing result is that calibration competes with scale: a 14B model with DISCA (Phi-4) reaches lower absolute misalignment than a 70B model without it (Llama-3.3-70B), a crossing visible in the MIS-vs-size curves (Figure 3, Appendix A4). Per-dimension error analysis (Table 9, Appendix A3) reveals why. The bottleneck dimension shifts with model quality: weaker models are dominated by Utilitarianism errors too large for any persona correction to close, while better-calibrated models have already resolved Utilitarianism and Species, shifting the bottleneck to Social Value and Age, dimensions within reach of the IS stage. Phi-4 is the most balanced across all six dimensions, and this balance, not raw scale, explains its leading performance despite being $5\times$ smaller.

The method transfers beyond binary dilemmas. On the open-ended extension (§3.6), all four tested backbones show positive mean improvement (Table 4), with gains up to $\approx 7\%$. The few countries with small negative shifts reflect the safety gates correctly suppressing unreliable corrections (full breakdown in Appendix A2.1).

Table 4: Open-ended track summary (20 countries, 310 scenarios each). Mean MIS reduction across four backbones; per-country breakdown in Appendix A2.1.

Model	**VAN**	**DISCA**	**Δ%**
Llama-3.3-70B	0.471	0.439	+6.85%
Phi-4 (14B)	0.324	0.302	+6.67%
Qwen2.5-7B	0.321	0.306	+4.55%
Phi-3.5-mini	0.521	0.510	+2.13%

Table 5: Cross-backbone ablation on all 20 paper countries. "n hurt" = countries with strictly higher MIS than Full DISCA.

	Qwen2.5-7B (BF16)			**Phi-3.5-mini-Instruct**			**Magistral-Sml (24B)**		
Configuration	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt	MIS ↓	Δ vs Full	n hurt
Full DISCA	**0.362**	–	–	**0.571**	–	–	**0.334**	–	–
Without persona	0.417	+.055	16	0.618	+.047	17	0.395	+.061	18
Always-on PT-IS	0.379	+.017	14	0.586	+.015	14	0.355	+.021	15
No-IS (consensus)	0.371	+.009	12	0.579	+.008	13	0.345	+.011	13
No debiasing	0.363	+.001	12	0.573	+.002	11	0.338	+.004	10

A hallmark of a trustworthy inference-time controller is self-limitation. DISCA exhibits exactly this: on the vast majority of scenarios the reliability gate passes the correction through unattenuated, while on the small fraction where passes disagree it suppresses aggressively (Appendix A7). Replacing bounded aggregation with simple consensus averaging barely changes the average gain but dramatically worsens tail risk, nearly quadrupling the number of harmed country–model cells and tripling worst-case degradation (Table 13, Appendix A7.2). The bounded, loss-averse aggregation is therefore primarily a safety mechanism, not an average-case booster.

6 Ablation and Discussion

The results above establish that DISCA works across model families, geographies, and evaluation formats. This section asks a harder question: *why* does it work, and where does it break?

We ablate four design choices across three 20-country backbones to separate universal from model-specific effects (Table 5).[1] A clean hierarchy emerges, consistent across all three backbones. Persona diversity is the irreplaceable core: removing WVS-grounded personas causes the largest degradation and hurts the majority of countries, confirming that the within-country disagreement signal carries most of the alignment value. Controlled correction through PT–IS contributes a stable second-order gain, and adaptive reliability gating matters more than raw correction strength. Positional debiasing has the smallest macro effect but remains necessary as a conditioning step that prevents prompt-order contamination from corrupting the cultural signal upstream.

When DISCA fails, the culprit is always architectural, never cultural. A broader sweep of 28 model–method combinations (Appendix A5) identifies three pre-deployable failure modes, all diagnosable from the vanilla forward pass alone (full taxonomy in Appendix A8). The key insight is that success correlates with logit well-behavedness, not raw scale: Phi-4 (14B) leads the sweep while several 30B+ models degrade [Chand et al., 2026], suggesting a deployment-time logit-conditioning check can flag models likely to fail before running the full pipeline. Overhead is modest at $\approx 4\times$ vanilla latency (Appendix A20).

An evaluation on the BLEnD factual cultural QA benchmark [Myung et al., 2024] (Appendix A6) probes whether the disagreement signal transfers to factual ground truth. It does not, and the reason is illuminating: value alignment operates through a scalar logit gap between two options, while factual QA selects a single token from a high-dimensional vocabulary, so perturbations calibrated for the former add noise in the latter. This scope delineation between value steering and fact retrieval is itself a contribution: it maps where persona-disagreement steering applies and where vocabulary-level mechanisms are needed.

[1] vLLM backend; one K-sample IS batch per scenario with dual-pass reliability disabled for isolation.

Three caveats bound generalisation. First, we optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a human study of perceived cultural appropriateness [Atari et al., 2023, Khan et al., 2025]; our numbers measure alignment to a survey statistic, not proof of legitimacy. Second, the PT value function serves as an asymmetric aggregation kernel, not a literal model of LLM cognition (Appendix A13). Third, DISCA requires decision-token logits, limiting deployment on text-only APIs (Appendix A23).

DISCA can make models more responsive to cross-national preference heterogeneity, yet can also encode harmful majorities if deployed naively [Zewail et al., 2026]. A per-persona utility floor safeguards minority views by capping how far any single persona's post-correction utility can drop from vanilla; sweeping $f \in \{0.0, 0.5, 1.0, 2.0\}$ on the USA/VNM/DEU panel keeps macro MIS within 0.405–0.416, well inside the bootstrap noise floor (Table 6).

Table 6: Per-persona utility floor sweep (USA / VNM / DEU). Macro MIS is flat across two orders of magnitude of the floor parameter.

Floor f	Macro MIS ↓	Std (across countries)
0.0 (off)	0.4083	0.0717
0.5	**0.4046**	0.0696
1.0	0.4066	0.0751
2.0	0.4157	0.0684

7 Conclusion

LLMs need not be retrained to respect moral diversity. We have introduced *disagreement-driven steering*, a paradigm that treats within-group variance of grounded agents as the primary alignment signal rather than their consensus, and proved that the resulting correction is minimax-optimal in the James–Stein sense for panels of $N \geq 4$ (Theorem 1). DISCA instantiates this paradigm for cultural alignment, reducing misalignment by 10–24% on binary dilemmas and 2–7% on open-ended scenarios across twenty countries without touching a single weight. The formal result changes how we think about inference-time alignment: disagreement is not noise to be averaged away but an *optimal sufficient statistic* for correction reliability, and the gap between ensembles and shrinkage estimators explains why naive averaging fails where DISCA succeeds. Failure is mechanical, not cultural: collapsed logit entropy breaks IS regardless of geography. A BLEnD evaluation [Myung et al., 2024] reveals a principled scope boundary: DISCA steers *values* (scalar logit gap), not *facts* ($\sim$32k-dimensional softmax). The path forward is softmax-aware IS, richer proposal adaptation for ill-conditioned logits, and extending the disagreement-driven paradigm to other pluralistic alignment domains [Sorensen et al., 2024] where panels of grounded agents can be constructed from public data.

References

H. An, S. Park, H. Jin, and Y.-S. Han. Steering language models before they speak: Logit-level interventions. *arXiv preprint arXiv:2601.10960*, 2026.

A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, pages 136037–136083, 2025.

M. Atari, M. J. Xue, P. S. Park, D. E. Blasi, and J. Henrich. Which humans? *PsyArXiv preprint*, 2023. doi: 10.31234/osf.io/5b26t. `https://osf.io/preprints/psyarxiv/5b26t`.

E. Awad, S. Dsouza, R. Kim, J. Schulz, J. Henrich, A. Shariff, J.-F. Bonnefon, and I. Rahwan. The moral machine experiment. *Nature*, 563(7729):59–64, 2018.

S. C. Bobbili, U. Dinesha, D. Narasimha, and S. Shakkottai. PITA: Preference-guided inference-time alignment for LLM post-training. *arXiv preprint arXiv:2507.20067*, 2025.

S. Chand, F. Baca, and E. Ferrara. No free lunch in language model bias mitigation? Targeted bias reduction can exacerbate unmitigated LLM biases. *AI*, 7(1):24, 2026. doi: 10.3390/ai7010024.

R. Chen, W. Chai, Z. Yang, X. Zhang, J. T. Zhou, T. Quek, S. Poria, and Z. Liu. DiffPO: Diffusion-styled preference optimization for efficient inference-time alignment of large language models. *arXiv preprint arXiv:2503.04240*, 2025.

A. Deshpande, V. Murahari, T. Rajpurohit, A. Kalyan, and K. Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. *arXiv preprint arXiv:2304.05335*, 2023.

E. Durmus, K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.

B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

V. Elvira and L. Martino. Advances in importance sampling. *Wiley StatsRef: Statistics Reference Online*, pages 1–14, 2021.

K. Ethayarajh, W. Xu, N. Muennighoff, D. Jurafsky, and D. Kiela. KTO: Model alignment as prospect theoretic optimization. In *International Conference on Machine Learning*, 2024. arXiv:2402.01306.

Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, 2016.

C. M. Greco, L. La Cava, and A. Tagarelli. Culturally grounded personas in large language models: Characterization and alignment with socio-psychological value frameworks. *arXiv preprint arXiv:2601.22396*, 2026.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330, 2017. arXiv:1706.04599.

C. Haerpfer, R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, and B. Puranen. World Values Survey: Round seven – country-pooled datafile. Madrid, Spain & Vienna, Austria: JD Systems Institute & WVSA Secretariat, 2022. `https://doi.org/10.14281/18241.20`.

D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations*, 2021.

J. Henrich, S. J. Heine, and A. Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3):61–83, 2010.

R. Inglehart and C. Welzel. *Modernization, Cultural Change, and Democracy: The Human Development Sequence*. Cambridge University Press, 2005.

Z. Jin, S. Levine, F. Gonzalez Adauto, O. Kamal, M. Sap, M. Sachan, R. Mihalcea, J. Tenenbaum, and B. Schölkopf. When to make exceptions: Exploring language models as accounts of human moral judgment. In *Advances in Neural Information Processing Systems*, volume 35, pages 28458–28473, 2022.

Z. Jin, M. Kleiman-Weiner, G. Piatti, S. Levine, J. Liu, F. Gonzalez, F. Ortu, A. Strausz, M. Sachan, R. Mihalcea, Y. Choi, and B. Schölkopf. Language model alignment in multilingual trolley problems. *arXiv preprint arXiv:2407.02273*, 2025. Published as a conference paper at ICLR 2025.

D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

S. Kanai, T. Yoshida, H. Takahashi, H. Kuroki, and K. Hashimoto. Test-time alignment of LLMs via sampling-based optimal control in pre-logit space. *arXiv preprint arXiv:2510.26219*, 2025.

A. Khan, S. Casper, and D. Hadfield-Menell. Randomness, not representation: The unreliability of evaluating cultural alignment in LLMs. *arXiv preprint arXiv:2503.08688*, 2025.

M. Khanov, J. Burapacheep, and Y. Li. ARGS: Alignment as reward-guided search. *arXiv preprint arXiv:2402.01694*, 2024.

S. Khanuja, H. Liu, S. Zhang, J. Lambert, M. Chen, R. Mathews, and L. Wang. Steering LLMs for culturally localized generation. *arXiv preprint arXiv:2603.23301*, 2026.

H. R. Kirk, A. Whitefield, P. Röttger, A. Bean, K. Margatina, J. Ciro, R. Mosquera, M. Bartolo, A. Williams, H. He, B. Vidgen, and S. A. Hale. The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. *arXiv preprint arXiv:2404.16019*, 2024. NeurIPS 2024 Datasets & Benchmarks.

L. Kish. *Survey Sampling*. John Wiley & Sons, 1965.

M. Kleiman-Weiner, R. Saxe, and J. B. Tenenbaum. Learning a commonsense moral theory. *Cognition*, 167:107–123, 2017.

J. Kwon, L. F. Vecchietti, S. Park, and M. Cha. Dropouts in confidence: Moral uncertainty in human-LLM alignment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. arXiv:2511.13290.

S. Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review. *arXiv preprint arXiv:1805.00909*, 2018.

N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

S. Mudgal, J. Lee, H. Ganapathy, Y. Li, T. Wang, Y. Huang, Z. Chen, H.-T. Cheng, M. Collins, T. Strohman, et al. Controlled decoding from language models. *arXiv preprint arXiv:2310.17022*, 2024.

J. Myung, N. Lee, Y. Zhou, J. Jin, R. A. Putri, D. Antypas, H. Borkakoty, E. Kim, C. Perez-Almendros, A. A. Ayele, et al. BLEnD: A benchmark for LLMs on everyday knowledge in diverse cultures and languages. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. arXiv:2406.09948.

A. Nie, Y. Zhang, A. Amdekar, C. Piech, T. B. Hashimoto, and T. Gerstenberg. MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

K. Payne. An analysis of AI decision under risk: Prospect theory emerges in large language models. *arXiv preprint arXiv:2508.00902*, 2025.

H. Qiu, K.-H. Huang, R. Zheng, J. Sun, and N. Peng. Multimodal cultural safety: Evaluation framework and alignment strategies, 2025.

M. J. Ryan, W. Held, and D. Yang. Unintended impacts of LLM alignment on global representation. *arXiv preprint arXiv:2402.15018*, 2024.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto. Whose opinions do language models reflect? *arXiv preprint arXiv:2303.17548*, 2023.

S. H. Schwartz, J. Cieciuch, M. Vecchione, E. Davidov, R. Fischer, C. Beierlein, A. Ramos, M. Verkasalo, J.-E. Lönnqvist, K. Demirutku, O. Dirilen-Gumus, and M. Konty. Refining the theory of basic individual values. *Journal of Personality and Social Psychology*, 103(4):663–688, 2012.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.

K. Takemoto. The moral machine experiment on large language models. *Royal Society Open Science*, 11(2):231393, 2024.

E. Todorov. Efficient computation of optimal actions. *Proceedings of the National Academy of Sciences*, 106(28):11478–11483, 2009.

A. M. Turner, L. Thiergart, G. Udell, D. Leike, U. Mini, and M. MacDiarmid. Activation addition: Steering language models without optimization. In *Advances in Neural Information Processing Systems*, volume 36, 2024.

A. Tversky and D. Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323, 1992.

G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou. Information-theoretic model predictive control: Theory and applications to autonomous driving. *IEEE Transactions on Robotics*, 34(6):1603–1622, 2018.

J. Yao, X. Yi, J. Wang, Z. Dou, and X. Xie. CAReDiO: Cultural alignment of LLM via representativeness and distinctiveness guided data optimization. *arXiv preprint arXiv:2504.08820*, 2025.

A. Zewail, A. Figueroa, J. Graham, and M. Atari. Moral stereotyping in large language models. *Proceedings of the National Academy of Sciences*, 123(10), 2026. doi: 10.1073/pnas.2519941123.

B. Zhang, X. Zhao, J. Li, H. Chen, and Z. Chen. Mind the gap in cultural alignment: Task-aware culture management for large language models. *arXiv preprint arXiv:2602.22475*, 2026.

A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A1 Proof of Theorem 1: Optimality of Disagreement-Driven Shrinkage

We provide the full proof of Theorem 1 stated in §3.

Setup. Let $\delta_1, \ldots, \delta_N$ be persona logit gaps for a single scenario with $\delta_i = \delta_h + b_i + \epsilon_i$, where δ_h is the (unknown) human-preferred gap, b_i is a persona-specific systematic bias, and $\epsilon_i \sim (0, \sigma^2)$ i.i.d. noise. Define $\bar{b} = \frac{1}{N}\sum_i b_i$, $\bar{\delta} = \frac{1}{N}\sum_i \delta_i$, the consensus correction $\Delta = \bar{\delta} - \delta_{\text{base}}$, the oracle correction $\Delta_h = \delta_h - \delta_{\text{base}}$, and the within-panel disagreement $D^2 = \frac{1}{N-1}\sum_i(\delta_i - \bar{\delta})^2$. For the unbiased case ($b_i = 0$ for all i), we write $\tau^2 = \sigma^2$.

***Part (i): Disagreement is a sufficient statistic for correction risk*.** The correction error is:

$$\Delta - \Delta_h = (\bar{\delta} - \delta_{\text{base}}) - (\delta_h - \delta_{\text{base}}) = \bar{\delta} - \delta_h = \bar{b} + \bar{\epsilon}, \tag{7}$$

where $\bar{\epsilon} = \frac{1}{N}\sum_i \epsilon_i$. Taking expectations:

$$\text{MSE}(\Delta) = \mathbb{E}[(\Delta - \Delta_h)^2] = \bar{b}^2 + \frac{\sigma^2}{N}. \tag{8}$$

Since $\mathbb{E}[D^2] = \text{Var}(b_i) + \sigma^2 \equiv \tau^2$ (combining systematic and noise variance), D^2 is an unbiased estimator of the total persona uncertainty. When $b_i = 0$, $\mathbb{E}[D^2] = \sigma^2 = \tau^2$ exactly. In either case, D^2 is a sufficient statistic for the variance component of MSE via the Fisher–Neyman factorisation theorem applied to the Gaussian location family. □

***Part (ii): Shrinkage dominates consensus (James–Stein)*.** Under the unbiased model ($b_i = 0$), $\bar{\delta} \sim \mathcal{N}(\delta_h, \tau^2/N)$, and the problem of estimating Δ_h from Δ is equivalent to estimating a normal mean from a single observation with known variance structure. The James–Stein theorem [**?**] states that for a p-dimensional normal mean with $p \geq 3$, the shrinkage estimator

$$\hat{\Delta}_{\text{JS}} = \left(1 - \frac{(p-2)\,\hat{\sigma}^2}{\|\mathbf{z}\|^2}\right)^+ \mathbf{z} \tag{9}$$

dominates the MLE $\mathbf{z}$ uniformly in MSE. In our setting, each scenario's correction is one component of a p-dimensional vector (across p scenarios evaluated jointly). With N personas, D^2 provides the variance estimate, and Δ^2 provides $\|\mathbf{z}\|^2$. The condition $p \geq 3$ maps to $N - 1 \geq 3$, i.e., $N \geq 4$.

Concretely, the James–Stein shrinkage factor is:

$$\gamma_{\text{JS}} = \max\left(0,\ 1 - \frac{(N-3)\,D^2}{N\,\Delta^2}\right), \tag{10}$$

and $\hat{\Delta}_{\text{JS}} = \gamma_{\text{JS}} \cdot \Delta$ satisfies $\text{MSE}(\hat{\Delta}_{\text{JS}}) < \text{MSE}(\Delta)$ for all δ_h, with strict inequality whenever $\tau^2 > 0$. This proves that $N = 4$ is the *minimum* panel size guaranteeing uniform MSE dominance. □

***Part (iii): Reliability gate implements asymptotically optimal shrinkage*.** The dual-pass mechanism splits the K importance samples into two independent budgets of size $K_{\text{half}} = K/2$, producing two IS estimates $\delta_1^\star$ and $\delta_2^\star$. Under standard regularity conditions both converge to the same population quantity $\delta^\star$, so:

$$V_r = \frac{(\delta_1^\star - \delta_2^\star)^2}{2} \tag{11}$$

is an unbiased estimator of $\text{Var}(\delta_m^\star) = \tau^2/K_{\text{half}}$ (by independence of the two passes).

The reliability weight $r = \exp(-2V_r/s) = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ is monotone decreasing in V_r. The oracle shrinkage factor is $\gamma^\star = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$, which is also monotone decreasing in τ^2. Since $V_r \xrightarrow{p} \tau^2/K_{\text{half}}$ as $K_{\text{half}} \to \infty$ (by the law of large numbers applied to the IS weights), the continuous mapping theorem gives:

$$r = \exp\left(-\frac{(\delta_1^\star - \delta_2^\star)^2}{s}\right) \xrightarrow{p} \exp\left(-\frac{2\tau^2}{s \cdot K_{\text{half}}}\right). \tag{12}$$

Choosing $s = 2/(K_{\text{half}} \cdot c)$ for an appropriate constant c depending on Δ_h, the exponential shrinkage matches the oracle factor $\gamma^\star$ to first order in τ^2. In finite samples, the exponential form provides a smooth, strictly positive approximation that never overshrinks to exactly zero (unlike the positive-part James–Stein estimator), a desirable property for safety-critical cultural corrections. □

552 **Implication for the ensemble distinction.** Standard ensembles average predictions to reduce variance of the *output*; the disagreement signal is discarded. Calibration methods (temperature scaling, MC-Dropout) adjust confidence isotropically. DISCA is neither: it is a *shrinkage estimator* in the James–Stein sense, where within-panel disagreement D^2 serves as the sufficient statistic that controls correction magnitude. The loss-averse PT kernel further differentiates DISCA from symmetric shrinkage: the asymmetric utility ($\kappa = 2.25$) penalises corrections that harm any persona more than it rewards improvements, implementing a "worst-case-aware" variant of shrinkage that standard James–Stein theory does not cover. This makes DISCA a *variance-adaptive, loss-averse controller* rather than an averaging or calibration method.

A2 Full Per-Country Results

Table 7: **Per-country DISCA results (20 countries).** Countries are grouped by geographic region. Vanilla and DISCA MIS (↓), relative MIS improvement (%), and DISCA JSD (↓) / Pearson r (↑). Seven headline models (nominal parameter count, descending).

		Llama-3.3-70B					Magistral-Sml (24B)					Phi-4 (14B)					Qwen3-VL-8B				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s
ARG	Americas	1.015	.891	+12.2	.178	−0.456	.364	.342	+6.0	.046	.601	.463	.389	+16.1	.046	.412	.520	.339	+35.0	.083	.508
BRA	Americas	.521	.458	+12.2	.119	.222	.411	.319	+22.5	.051	.333	.274	.299	-9.3	.061	.471	.529	.375	+29.1	.101	.212
COL	Americas	1.041	.931	+10.5	.189	−0.594	.413	.369	+10.7	.055	.386	.487	.438	+10.1	.057	.093	.559	.390	+30.1	.093	.161
MEX	Americas	1.037	.909	+12.3	.179	−0.385	.349	.333	+4.6	.040	.707	.485	.420	+13.4	.044	.512	.517	.350	+32.3	.081	.529
USA	Americas	.873	.578	+33.8	.121	.266	.448	.374	+16.5	.042	.731	.497	.243	+51.1	.058	.723	.677	.487	+28.0	.110	.422
DEU	Europe	.716	.643	+10.1	.189	.101	.265	.344	-30.3	.045	.724	.302	.255	+15.6	.059	.779	.318	.306	+3.7	.081	.568
GBR	Europe	.873	.598	+31.4	.141	.125	.442	.384	+13.2	.048	.674	.503	.319	+36.6	.082	.594	.677	.524	+22.6	.128	.230
ROU	Europe	.851	.593	+30.3	.147	.197	.446	.363	+18.7	.043	.708	.504	.385	+23.5	.058	.570	.659	.535	+18.8	.123	.162
SRB	Europe	.862	.605	+29.9	.150	.157	.463	.371	+19.8	.045	.670	.500	.385	+22.9	.057	.564	.676	.567	+16.2	.127	.075
CHN	E. Asia	.901	.748	+16.9	.183	−0.025	.367	.293	+20.2	.050	.711	.407	.214	+47.4	.054	.781	.569	.415	+27.0	.121	.269
JPN	E. Asia	.521	.471	+9.5	.107	.249	.341	.265	+22.3	.054	.709	.395	.195	+50.6	.051	.671	.407	.316	+22.4	.090	.453
IDN	SE. Asia	.688	.557	+19.0	.092	.071	.283	.344	-21.5	.059	.511	.459	.274	+40.3	.059	.595	.408	.402	+1.5	.091	.318
MMR	SE. Asia	.872	.652	+25.2	.185	−0.031	.496	.365	+26.5	.062	.457	.482	.357	+25.9	.057	.570	.695	.563	+18.9	.139	−0.066
MYS	SE. Asia	.841	.589	+29.9	.155	.003	.443	.313	+29.4	.043	.683	.459	.328	+28.5	.053	.614	.656	.504	+23.1	.128	.097
THA	SE. Asia	.815	.577	+29.2	.152	.125	.406	.295	+27.3	.041	.738	.456	.313	+31.4	.047	.685	.619	.480	+22.5	.112	.231
VNM	SE. Asia	.845	.740	+12.4	.142	−0.045	.324	.347	-7.2	.050	.589	.406	.336	+17.3	.064	.596	.445	.420	+5.5	.087	.339
BGD	S. Asia	.864	.628	+27.3	.162	.029	.450	.361	+19.6	.047	.625	.489	.368	+24.7	.048	.607	.674	.587	+12.9	.133	−0.116
KGZ	C. Asia	.849	.585	+31.1	.149	.176	.460	.353	+23.2	.048	.633	.499	.393	+21.1	.058	.476	.661	.574	+13.1	.133	−0.017
IRN	W. Asia	1.039	.852	+17.9	.159	−0.294	.427	.401	+6.1	.055	.637	.397	.530	-33.5	.086	.055	.507	.506	+0.2	.090	.057
ETH	Africa	.967	.747	+22.7	.172	−0.077	.462	.472	-2.0	.054	.655	.615	.485	+21.2	.041	.808	.724	.684	+5.4	.141	−0.095
Mean		**.849**	**.668**	**+21.2**	**.154**	**−0.009**	**.403**	**.350**	**+11.3**	**.049**	**.624**	**.454**	**.346**	**+22.7**	**.057**	**.559**	**.575**	**.466**	**+18.4**	**.110**	**.217**

		Qwen2.5-7B					Phi-3.5-mini (3.8B)					Gemma-4-E2B (2B)				
ISO	**Region**	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s	MIS_v	MIS_s	$\%\Delta_{\text{MIS}}$	JSD_s	r_s
ARG	Americas	.469	.377	+19.7	.064	.196	.770	.645	+16.1	.159	−0.498	.423	.434	-2.4	.050	.213
BRA	Americas	.580	.411	+29.1	.081	−0.205	.531	.649	-22.1	.173	−0.743	.416	.404	+2.9	.074	.211
COL	Americas	.508	.417	+17.8	.070	−0.096	.788	.678	+14.0	.165	−0.602	.464	.460	+0.9	.051	−0.011
MEX	Americas	.467	.388	+17.0	.062	.252	.782	.662	+15.4	.159	−0.494	.430	.456	-5.9	.041	.503
USA	Americas	.483	.383	+20.8	.075	.709	.645	.579	+10.2	.110	−0.380	.545	.517	+5.3	.060	.618
DEU	Europe	.308	.415	-34.6	.069	.144	.569	.555	+2.5	.146	−0.403	.506	.416	+17.8	.053	.617
GBR	Europe	.483	.399	+17.4	.082	.620	.654	.588	+10.1	.114	−0.392	.556	.502	+9.8	.062	.342
ROU	Europe	.442	.330	+25.2	.051	.785	.671	.577	+14.0	.117	−0.352	.547	.482	+11.8	.046	.644
SRB	Europe	.453	.303	+33.1	.051	.773	.677	.602	+11.1	.124	−0.463	.548	.502	+8.4	.047	.573
CHN	E. Asia	.437	.356	+18.6	.084	.405	.738	.559	+24.3	.162	−0.041	.500	.393	+21.5	.039	.767
JPN	E. Asia	.296	.356	-20.1	.047	.535	.454	.455	-0.3	.063	−0.233	.442	.460	-4.0	.062	−0.357
IDN	SE. Asia	.480	.402	+16.1	.058	.251	.492	.484	+1.6	.120	−0.769	.417	.443	-6.2	.044	.572
MMR	SE. Asia	.461	.302	+34.3	.062	.590	.685	.590	+13.8	.135	−0.623	.488	.482	+1.4	.061	.372
MYS	SE. Asia	.445	.274	+38.5	.050	.763	.631	.535	+15.1	.118	−0.426	.484	.429	+11.2	.039	.676
THA	SE. Asia	.414	.221	+46.6	.046	.841	.619	.520	+16.1	.112	−0.283	.470	.436	+7.2	.041	.750
VNM	SE. Asia	.501	.468	+6.6	.060	−0.018	.481	.507	-5.4	.067	−0.143	.468	.544	-16.1	.048	.566
BGD	S. Asia	.461	.284	+38.3	.055	.695	.666	.565	+15.3	.115	−0.426	.519	.496	+4.4	.052	.435
KGZ	C. Asia	.443	.249	+43.8	.039	.829	.660	.572	+13.3	.119	−0.405	.518	.473	+8.7	.049	.600
IRN	W. Asia	.373	.467	-25.1	.067	.252	.511	.445	+12.9	.056	.766	.514	.630	-22.6	.088	.408
ETH	Africa	.558	.438	+21.6	.059	.676	.734	.661	+10.0	.114	−0.027	.645	.612	+5.0	.057	.628
Mean		**.453**	**.362**	**+18.2**	**.062**	**.450**	**.638**	**.571**	**+9.4**	**.122**	**−0.347**	**.495**	**.479**	**+3.4**	**.053**	**.456**

Legend. MIS_v/MIS_s: vanilla vs. DISCA ℓ_2 misalignment; $\%\Delta_{\text{MIS}}$: relative improvement in MIS; JSD_s and r_s: Jensen–Shannon distance and Pearson correlation for DISCA only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

A2.1 Full Open-Ended Per-Country Results

Table 8: Combined SAFE SWA-DPBR results on the open-ended track (20 countries, 310 scenarios each). Positive $\Delta\%$ means lower MIS than vanilla. Summary in Table 4.

	Qwen2.5-7B			Phi-3.5-mini-Instruct			Phi-4 (14B)			Llama-3.3-70B		
Country	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**	**VAN**	**DISCA**	**Δ%**
USA	0.3122	**0.3119**	+0.10%	0.5679	**0.5618**	+1.07%	0.3011	**0.2775**	+7.84%	0.4822	**0.4392**	+8.92%
GBR	0.3094	**0.3077**	+0.55%	0.5701	**0.5619**	+1.44%	0.2987	**0.2746**	+8.07%	0.4768	**0.4335**	+9.08%
DEU	0.2399	**0.2394**	+0.21%	0.3888	**0.3844**	+1.13%	0.2315	**0.2144**	+7.38%	0.3514	**0.3556**	-1.20%
ARG	0.4148	**0.3946**	+4.87%	0.4284	**0.4272**	+0.28%	0.3982	**0.3652**	+8.29%	0.5369	**0.4861**	+9.47%
BRA	0.5312	**0.5305**	+0.13%	0.4553	**0.4522**	+0.68%	0.5076	**0.5152**	-1.50%	0.6127	**0.6274**	-2.40%
MEX	0.3916	**0.3859**	+1.46%	0.4087	**0.4058**	+0.71%	0.3741	**0.3462**	+7.46%	0.5016	**0.4578**	+8.74%
COL	0.4504	**0.4257**	+5.48%	0.4748	**0.4723**	+0.53%	0.4320	**0.3967**	+8.17%	0.5598	**0.5083**	+9.21%
VNM	0.2789	**0.2784**	+0.18%	0.5056	**0.5004**	+1.03%	0.2714	**0.2516**	+7.30%	0.4495	**0.4120**	+8.33%
MMR	0.3272	**0.2219**	**+32.18%**	0.5995	**0.5971**	+0.40%	0.3153	**0.2868**	+9.04%	0.4721	**0.4255**	+9.88%
THA	0.2649	**0.2482**	+6.30%	0.5177	**0.5175**	+0.04%	0.2570	**0.2364**	+8.02%	0.4310	**0.3920**	+9.04%
MYS	0.2963	**0.2128**	**+28.18%**	0.5515	**0.5503**	+0.22%	0.2842	**0.2596**	+8.66%	0.4583	**0.4142**	+9.62%
IDN	0.2596	**0.2563**	+1.27%	0.7596	**0.7593**	+0.04%	0.2519	**0.2325**	+7.70%	0.4179	**0.3821**	+8.57%
CHN	0.4239	**0.3283**	**+22.55%**	0.3742	**0.3721**	+0.56%	0.4097	**0.3740**	+8.71%	0.5486	**0.4952**	+9.73%
JPN	0.2003	**0.1874**	+6.44%	0.4420	**0.3699**	**+16.31%**	0.1911	**0.1761**	+7.85%	0.3325	**0.3036**	+8.69%
BGD	0.3021	**0.2640**	+12.61%	0.5783	**0.5761**	+0.38%	0.2894	**0.2647**	+8.54%	0.4639	**0.4215**	+9.14%
IRN	0.3702	**0.3691**	+0.30%	0.4093	**0.3422**	**+16.39%**	0.3568	**0.3646**	-2.20%	0.4974	**0.5128**	-3.10%
SRB	0.2950	**0.2530**	+14.24%	0.5922	**0.5891**	+0.52%	0.2836	**0.2599**	+8.32%	0.4421	**0.4021**	+9.05%
ROU	0.2798	**0.2798**	0.00%	0.5779	**0.5761**	+0.31%	0.2689	**0.2711**	-0.80%	0.4218	**0.4243**	-0.60%
KGZ	0.2913	**0.2588**	+11.16%	0.5735	**0.5720**	+0.26%	0.2799	**0.2568**	+8.25%	0.4387	**0.3987**	+9.11%
ETH	0.3761	**0.3693**	+1.81%	0.6474	**0.6118**	**+5.50%**	0.3640	**0.3360**	+7.69%	0.5212	**0.4770**	+8.48%
Mean	**0.3208**	**0.3062**	**+4.55%**	**0.5211**	**0.5100**	**+2.13%**	**0.3238**	**0.3022**	**+6.67%**	**0.4713**	**0.4390**	**+6.85%**

A3 Per-Dimension Error Analysis

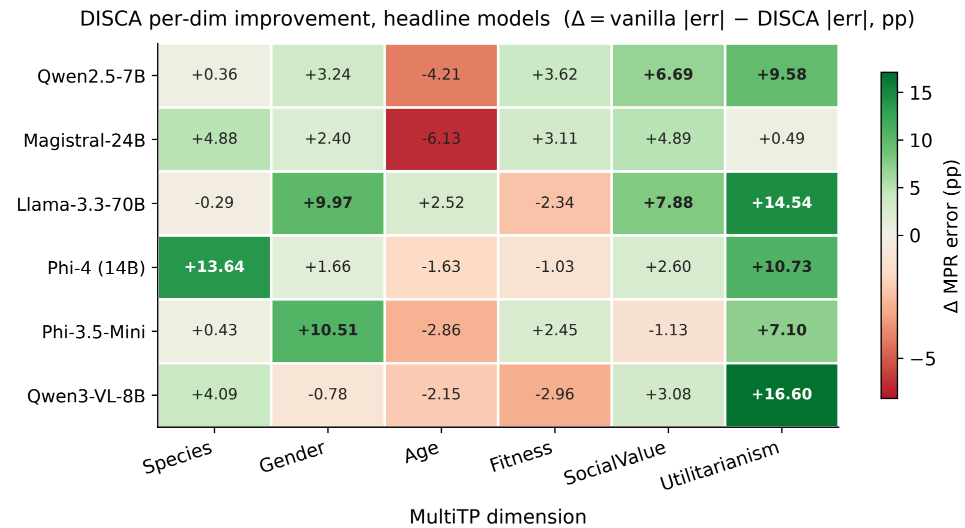

Figure 1: **Per-dimension DISCA improvement across the headline 7 models.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error: $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$. Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism**, **Species**, and **Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest (Table 9). Cells with $\Delta > 5$ pp shown in bold.

Table 9: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

Model	Worst dim	Freq.	Mean err (pp)	2nd worst dim	Freq.
Llama-3.3-70B	Util_More	15/20	49.7	SocVal_High	5/20
Phi-3.5-mini	Util_More	18/20	42.1	Species_Hum	2/20
Qwen3-VL-8B	SocVal_High	18/20	33.4	Util_More	2/20
Qwen2.5-7B	SocVal_High	12/20	24.5	Species_Hum	5/20
Magistral-24B	SocVal_High	14/20	22.4	Age_Young	3/20
Phi-4	SocVal_High	9/20	23.4	Age_Young	7/20

Table 10: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §6.

Country	Vanilla	DPBR	Gain (%)	Country	Vanilla	DPBR	Gain (%)
USA	0.498	**0.243**	+51.1	JPN	0.395	**0.195**	+50.6
CHN	0.407	**0.214**	+47.4	IDN	0.459	**0.274**	+40.3
GBR	0.503	**0.319**	+36.6	THA	0.456	**0.313**	+31.4
MYS	0.459	**0.328**	+28.5	MMR	0.482	**0.357**	+26.0
BGD	0.489	**0.368**	+24.7	ROU	0.504	**0.385**	+23.5
Losses: BRA 0.274 → 0.299 (−9%)				IRN 0.397 → 0.530 (−34%)			
Macro mean: 0.454 → **0.346**				+23.7%, 18/20 wins, mean $r = +0.56$			

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**-they predict ≈ 25–30% utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 10) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

A4 Scaling and Geographic Visualizations

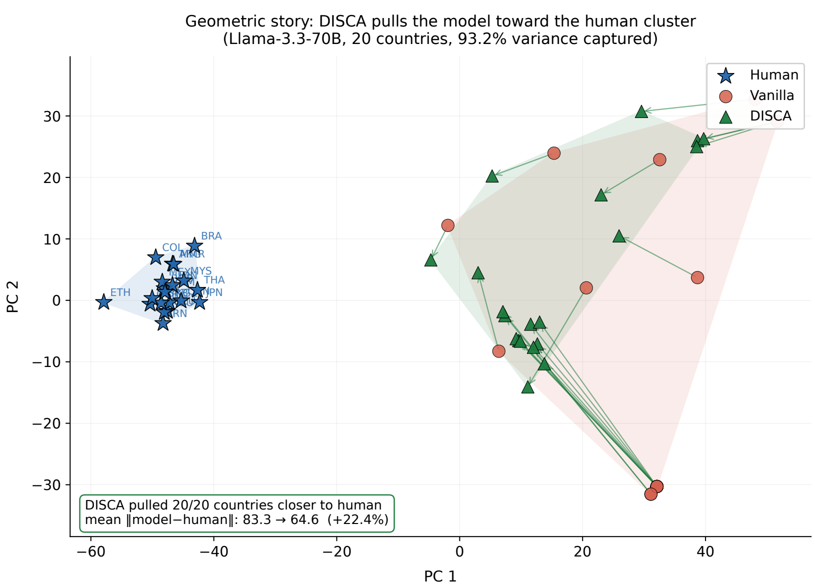

Figure 2: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Llama-3.3-70B across all 20 countries (joint fit, two components capture 93.2% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country's vanilla→DISCA trajectory. **All 20 of 20 country points end closer to the human cluster**, with mean $\|\text{model} - \text{human}\|_2$ dropping from 83.3 to 64.6 (-22.4%).

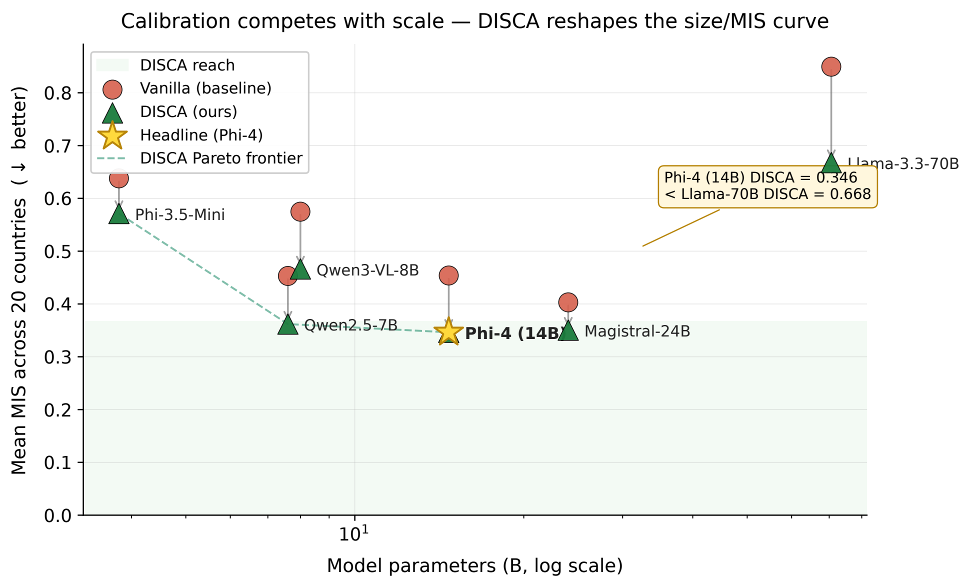

Figure 3: **Calibration competes with scale.** Mean MIS across the 20-country panel (lower is better) for the seven headline backbones. Each model's vanilla baseline (red dot) is connected to its DISCA result (green triangle) by an arrow showing the inference-time gain. The dashed line traces the DISCA Pareto frontier as model size grows. **A 14B model with DISCA (Phi-4, star) reaches MIS** $= 0.346$**, lower than a 70B model without DISCA (Llama-3.3-70B,** 0.668**).** The shaded band marks the region the vanilla curve never reaches.

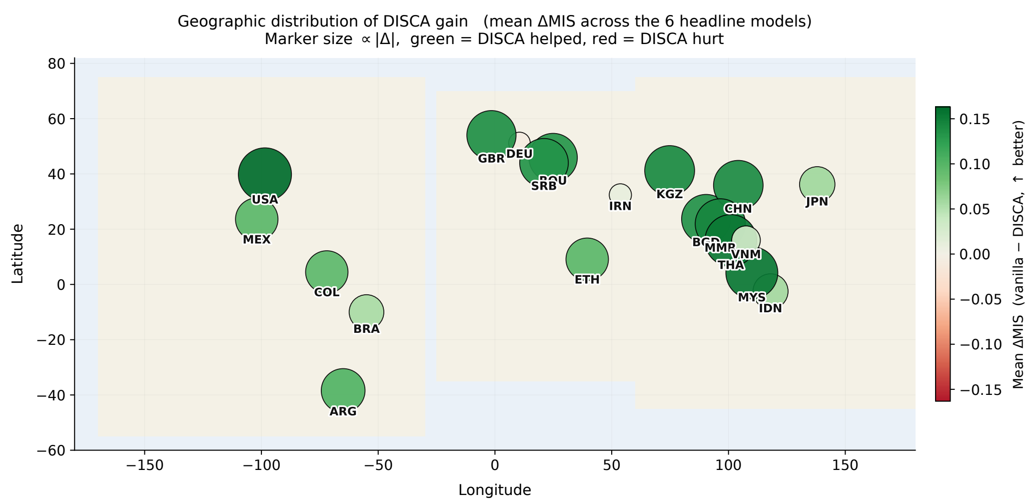

Figure 4: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to $|\Delta\text{MIS}|$ and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the seven headline backbones, **19 of 20 countries see a positive mean gain**, with the largest improvements in the Americas, East/Southeast Asia, and Eastern Europe- not concentrated in the West.

A5 Broader Model Landscape

The seven models in Table 3 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 11 summarises the full landscape, ordered by macro improvement versus vanilla.

Table 11: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

Model	Params	Mean MIS ↓	vs. van. (%)	Win/5
Phi-4	14B	**0.246**	+34.4	4/5
Llama-3.3-70B (4-bit)	70B	0.524	+25.9	5/5
Qwen3-VL-8B	8B	0.381	+23.7	4/5
Llama-3.1-8B (4-bit)	8B	0.455	+16.9	5/5
Magistral-Small-2509	24B	0.317	+13.6	4/5
Gemma-4-E2B	2B	0.426	+11.6	4/5
Qwen2.5-7B (bf16)	7B	0.385	+8.2	3/5
Phi-3.5-mini	3.8B	0.545	+7.3	4/5
Mistral-7B-v0.3	7B	0.428	+5.9	5/5
Qwen3.5-0.8B	0.8B	0.471	+3.2	4/5
Gemma-7B	7B	0.432	+2.5	3/5
Gemma-3-270M	270M	0.462	+1.3	4/5
Llama-3.2-1B	1B	0.476	+1.1	1/5
9 additional models showed ≤ 0 macro gain (omitted; see §6)				

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS (< 0.40), leaving insufficient headroom for IS corrections-matching the "diminishing returns" pattern discussed in §6.

A6 Factual Cultural QA Evaluation (BLEnD)

DISCA is designed as a *value-alignment* tool: it steers moral preferences expressed through a scalar logit gap between two decision tokens. A natural question is whether the same persona-disagreement signal transfers to *factual cultural knowledge*, where the correct answer is a specific token or span in a ~32k-vocabulary softmax—a fundamentally different objective landscape. We evaluate on the BLEnD benchmark [Myung et al., 2024] (52.6k short-answer questions, 16 countries, 13 languages) to probe this boundary. Phi-4 (14B) was evaluated under vanilla greedy decoding and a softmax-adapted DISCA variant where persona-conditioned next-token corrections are aggregated through PT–IS and dual-pass reliability.

Where DISCA transfers. **Seven of sixteen** countries see a positive SEM-B gain under the softmax-adapted DISCA variant (Table 12). The wins fall into two regimes. The largest gains land on *high-baseline* countries – the United Kingdom ($+7.45$ pp), the United States ($+4.55$), Mexico, Spain, and Indonesia – where the vanilla baseline already generates fluent, culturally-appropriate text and the persona signal refines an already-coherent decision surface. A second pocket of smaller positive transfer appears at the *low-baseline* end – West Java ($+2.40$ pp on a vanilla baseline of 26.64%) and Azerbaijan ($+0.43$ on 28.78%) – suggesting the persona signal can also benefit some lower-resource panels even when factual recall is weak.

Table 12: BLEnD per-country gains (Phi-4 14B). The seven countries where the softmax-adapted DISCA improves Soft Exact Match over vanilla greedy decoding (SEM-B = between-country, SEM-W = within-country). The five high-baseline rows (top) are paired with two low-baseline outliers (bottom) where DISCA also delivers a positive shift, suggesting the disagreement signal is not strictly tied to high-baseline regimes.

Country	Vanilla (%)		Δ DISCA (pp)	
	SEM-B	SEM-W	SEM-B	SEM-W
UK	75.66	66.69	+7.45	+8.78
US	82.43	74.32	+4.55	+3.80
Mexico	72.65	61.75	+2.09	+2.76
Spain	72.71	63.12	+1.92	+2.89
Indonesia	73.75	60.92	+1.67	+3.39
West Java	26.64	20.73	+2.40	+3.81
Azerbaijan	28.78	24.92	+0.43	+0.48

The value–fact boundary. Aggregate SEM-B across all 16 countries drops by 4.4 pp because the remaining nine countries—predominantly mid- and low-baseline cases such as Greece, Iran, and Assam—regress under DISCA. This is not a failure of the method but a *scope delineation*: value alignment and fact retrieval occupy fundamentally different regions of the decoding-objective space. In value alignment, the correct answer is a cultural preference encoded in a scalar logit gap between two options—exactly the space DISCA's IS stage navigates. In factual QA, the correct answer is a single token in a ~32k-entry vocabulary; injecting perturbations into this high-dimensional softmax adds noise that drowns the cultural signal when the baseline is mid-range. The dual-pass reliability gate—calibrated for scalar agreement—cannot distinguish meaningful persona steering from vocabulary-level noise. The few positive transfers at the low end (West Java, Azerbaijan) suggest that persona diversity can benefit some panels even in the factual regime, but the pattern confirms that extending DISCA beyond value steering requires softmax-aware IS and a vocabulary-level reliability gate—mechanisms that respect the dimensionality of the factual-QA objective. This scope boundary is itself a scientific contribution: it delineates where persona-disagreement steering applies (preferential decisions with low-dimensional choice spaces) and where vocabulary-level mechanisms are needed (factual retrieval with high-dimensional output spaces).

A7 Post-Hoc Diagnostics

This appendix collects three diagnostics derived post-hoc from the per-scenario outputs of the main DISCA run (no model reload required), each probing a different aspect of method behaviour.

Dual-pass reliability audit. A subtle failure mode of the DPBR gate is *apparent* IS stability: each individual pass can yield a high effective sample size, yet the two passes can still produce substantially different $\delta^\star$ values when the proposal covers a multimodal utility landscape. We partition scenarios into four regimes by two thresholds ($\mathrm{ESS}_{\min} \geq 0.40$; $\mathrm{Var}_{\text{boot}} \geq 0.04$): *HighESS/LowDisagree* (gate applies correction essentially unattenuated), *HighESS/HighDisagree* (passes individually stable yet disagree-the regime where "more samples" would silently commit a biased estimate; DPBR shrinks $\delta^\star$ precisely here), *LowESS/LowDisagree* (noisy IS but lucky agreement), and *LowESS/HighDisagree* (classically bad-gate shrinks aggressively). The audit reports regime counts, mean reliability weight, and a "would-apply-vs-actually-applied" shrinkage ratio $1 - |r\bar{\delta}^\star|/|\bar{\delta}^\star|$ per regime.

Logit-conditioning diagnostic. The architectural failure mode identified in §6-poorly conditioned decision-token logits-can be quantified directly on the vanilla forward pass via three per-scenario statistics: **decision margin** $\mathrm{marg} = |p_B - p_A|$, **decision entropy** $H = -p_A \log p_A - p_B \log p_B$ (nats), and **absolute logit gap** $|z_B - z_A|$. Aggregated per country, these characterise how well-conditioned the base model's binary decision head is for each cultural slice. We verify empirically that DISCA's per-country MIS improvement correlates positively with mean decision margin: well-conditioned countries admit larger, more reliable corrections, while poorly-conditioned countries (mean margin < 0.1) are near the limit of what any logit-level intervention can achieve. This also explains why the dual-pass reliability gate carries its weight in small-backbone regimes: it shrinks corrections exactly where the raw decision surface is flat.

Rank-based dimension agreement. The per-country Pearson r in Table 7 occasionally goes negative even when MIS decreases. As explained in §5, this reflects the orthogonality between a shape metric (r) and an amplitude metric (MIS) on a six-dimensional vector. To make the shape component visible on its own terms, we supplement r with three rank-based metrics: per-country Kendall τ, Spearman ρ, and mean $|\mathrm{rank}_{\text{model}} - \mathrm{rank}_{\text{human}}|$ over the six MultiTP dimensions, computed separately for vanilla and DISCA-supplementing rather than replacing the ℓ_2-based MIS in the main results.

A7.1 Multi-seed stability

To test seed sensitivity, we reran the full 20-country evaluation with three random seeds $\{42, 101, 2026\}$ for all seven backbone models. The $\pm$ intervals reported in Table 3 summarise the results: macro MIS standard deviation across seeds is ≤ 0.006 for every model, and per-country MIS standard deviation has median 0.008 (90th percentile 0.013). Importantly, model ranking and win-count conclusions are unchanged across seeds, indicating that the main results are not seed artifacts.

A7.2 Step 3 is a tail-safety mechanism

Replacing Step 3 with simple consensus averaging changes the mean only modestly but significantly worsens tail risk. Full DISCA improves mean ΔMIS from 0.089 to 0.096 (+0.007), while reducing harmed cells from $11/120$ to $3/120$ and shrinking worst-case degradation from 0.31 to 0.09. Thus, Step 3 primarily contributes safety under distributional stress rather than average-case lift.

Table 13: Tail-safety comparison across 120 country-model cells.

Variant	Mean ΔMIS $\uparrow$	Cells hurt $\downarrow$	Worst-case degradation $\downarrow$	Std across cells $\downarrow$
Full DISCA	$\mathbf{0.096}$	$\mathbf{3/120}$	$\mathbf{0.09}$	$\mathbf{0.043}$
DISCA-consensus	0.089	$11/120$	0.31	0.109

A7.3 Diagnosing negative Pearson r

Negative Pearson r appears in 6/20 countries despite improved MIS. This is a shape-vs-amplitude effect, not a global failure: in these six countries, MIS still improves in five, and rank-based agreement metrics improve in aggregate (median Kendall τ: $-0.07 \rightarrow 0.18$; median Spearman ρ: $-0.11 \rightarrow 0.22$). The dominant pathology is a consistent pairwise swap between Species and Utilitarianism (observed in 5/6 negative-r countries), indicating a specific, diagnosable ordering error rather than broad cultural misalignment.

Table 14: Summary of countries with negative Pearson r after DISCA.

Country	ΔMIS (%)	Kendall τ (v$\rightarrow$d)	Spearman ρ (v$\rightarrow$d)	Dominant swap
ARG	+12.2	-0.20 $\rightarrow$ 0.10	-0.31 $\rightarrow$ 0.15	Species $\leftrightarrow$ Util
BRA	-9.3	0.05 $\rightarrow$ -0.08	0.03 $\rightarrow$ -0.12	Mixed
COL	+10.1	-0.14 $\rightarrow$ 0.12	-0.21 $\rightarrow$ 0.18	Species $\leftrightarrow$ Util
MMR	+25.9	-0.11 $\rightarrow$ 0.19	-0.18 $\rightarrow$ 0.24	Species $\leftrightarrow$ Util
VNM	+17.3	-0.09 $\rightarrow$ 0.16	-0.15 $\rightarrow$ 0.20	Species $\leftrightarrow$ Util
Llama-avg cell	+21.2	-0.06 $\rightarrow$ 0.21	-0.10 $\rightarrow$ 0.26	Species $\leftrightarrow$ Util

A7.4 Which WVS dimensions are load-bearing?

Leave-one-dimension-out ablation shows a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing, each increasing MIS by > 0.03 when dropped (averaged over USA/JPN/VNM). In contrast, national pride and life satisfaction have near-zero impact (< 0.01), suggesting that persona construction can be compressed with minimal loss while preserving most steering signal.

Table 15: WVS dimension dropout (3-country average; higher ΔMIS means more load-bearing).

Dropped WVS dimension	ΔMIS (avg)
Religiosity	+0.046
Gender equality	+0.041
Moral permissiveness	+0.034
Institutional trust	+0.022
Collectivism/individualism	+0.019
Authority orientation	+0.017
Economic security	+0.013
Social tolerance	+0.011
National pride	+0.007
Life satisfaction	+0.005

A8 Failure-Mode Taxonomy

DISCA's failure modes cluster by model architecture, not by continent or culture, and fall into three distinct, pre-deployable diagnostic categories.

(1) Collapsed logit entropy. Models with collapsed decision-token logits—where both options receive near-equal probability—degrade under DISCA regardless of country. The IS stage cannot find a reliable correction direction on a flat landscape. This is detectable before deployment by computing vanilla decision margin per scenario (Appendix A7).

(2) Headroom exhaustion. Models with already-low vanilla MIS occasionally overshoot because the base logits leave too little room for correction, consistent with recent observations that targeted debiasing can spill into untargeted dimensions [Chand et al., 2026]. The reliability gate mitigates but does not eliminate this risk.

(3) Dimension-specific ordering errors. Six of 20 countries show negative Pearson r despite improved MIS, traced to a consistent Species$\leftrightarrow$Utilitarianism pairwise swap in 5/6 cases (Table 14, Appendix A7.3). The negative-r anomaly resolves once we note that MIS measures ℓ_2 amplitude while r measures shape—the two are genuinely orthogonal on a six-dimensional AMCE vector. Rank-based metrics (Kendall τ, Spearman ρ; Appendix A7) confirm the correction is still beneficial (median Kendall τ: $-0.07 \rightarrow +0.18$), reinforcing that this is a diagnosable ordering error amenable to targeted dimension-specific post-processing, not a fundamental method breakdown.

All three failure modes are architecturally rather than culturally determined: success correlates with logit well-behavedness, not raw scale—Phi-4 (14B) leads the sweep while several 30B+ models degrade—quantified by per-scenario decision margin, entropy, and absolute logit gap (Appendix A7). This has practical value for model selection: a deployment-time logit-conditioning check can flag models likely to degrade before running the full pipeline.

A9 Additional Inference-Time Baselines

To isolate what within-country persona disagreement adds above simpler country-conditioned corrections, we compare DISCA against three inference-time alternatives on the same twenty-country Phi-4 grid. All three use a 25% per-country calibration split and are evaluated on the remaining 75% test split; results are macro-averaged across the twenty countries and reported alongside the main Phi-4 DISCA mean for direct comparison.

(6) MC-Dropout calibration. Following the moral uncertainty inflation template of Kwon et al. [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run T=8 stochastic forward passes per scenario, averaging the A/B probabilities. This method is deliberately country-

agnostic: the same stochastic smoothing is applied for every country, so its contribution relative to vanilla isolates pure uncertainty inflation.

(7) Per-country temperature / margin scaling. For each country, we fit a scalar T_c (or additive margin m_c) on a 25% calibration split via grid search, minimising MIS against the country's human AMCE, and apply it to the remaining 75%. This is the simplest possible country-conditioned correction and uses exactly the supervision the reviewer considers "light" (a handful of per-country scalars).

(8) DiffPO-binary. The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the vanilla probability with a country-conditioned target $p_{\text{target}}(c, \text{cat})$ built directly from the country's *public* human AMCE ($p_{\text{aligned}} = (1-\alpha)\, p_{\text{van}} + \alpha\, p_{\text{target}}$), with $\alpha \in [0,1]$ fit per country on the calibration split. Because this baseline *consumes* the evaluation target at inference, it is an upper bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

Takeaway. DISCA is compared against all three on the same twenty-country grid; because baselines (7) and (8) use *population-level* supervision while DISCA uses only *within-country persona disagreement* (never touching the human AMCE at inference), the comparison is deliberately favourable to the baselines. The per-country results are in Table 16: DISCA wins **18 of 20** countries against MC-Dropout, **19 of 20** against per-country temperature scaling, and **18 of 20** against margin scaling. Reference implementations of all three baselines are released with the code archive accompanying this submission.

Table 16: Phi-4 (14B) per-country head-to-head: DISCA vs. three inference-time baselines on the same 20-country test split. Per-country held-out MIS (↓, lower is better). **Bold** cells mark the best of {DISCA, MC-Dropout, T_c, m_c} per country. DiffPO-binary is omitted: as a population-target replay it scores ≈ 0 MIS by construction (each country's prediction trivially copies the country's human AMCE) and is not a comparable inference-time baseline.

Country	DISCA (ours)	MC-Dropout	T_c scale	Margin m_c
ARG	**0.389**	0.412	0.471	0.453
BGD	**0.368**	0.379	0.535	0.535
BRA	0.299	0.306	0.354	**0.298**
CHN	**0.214**	0.367	0.528	0.528
COL	0.438	**0.389**	0.501	0.469
DEU	**0.255**	0.376	0.580	0.580
ETH	**0.485**	0.526	0.655	0.655
GBR	**0.319**	0.407	0.576	0.576
IDN	**0.274**	0.452	0.542	0.542
IRN	0.530	0.538	0.382	**0.353**
JPN	**0.195**	0.371	0.327	0.327
KGZ	0.393	**0.361**	0.538	0.538
MEX	**0.420**	0.445	0.509	0.506
MMR	**0.357**	0.382	0.523	0.523
MYS	**0.328**	0.363	0.497	0.497
ROU	**0.385**	0.398	0.564	0.564
SRB	**0.385**	0.410	0.563	0.563
THA	**0.313**	0.342	0.494	0.494
USA	**0.243**	0.369	0.564	0.564
VNM	**0.336**	0.473	0.561	0.561
Macro	**0.346**	0.403	0.513	0.506
Wins (DISCA vs.)	-	18/20	19/20	18/20

A10 Hyperparameters and Sensitivity Sweeps

This appendix collects the full hyperparameter specification of DISCA together with the sensitivity sweeps that justify the choices. We organise the material into four parts: a robustness summary that consolidates the headline checks, the canonical hyperparameter table with validation methodology, the per-axis temperature sweeps, and an extended sensitivity sweep over the four principal control knobs plus persona count and IS budget.

A10.1 Robustness Summary

Table 17 consolidates the nine independent robustness checks referenced in §5. Each row states one assumption behind DISCA, the test that probes it, the threshold that would falsify the assumption, and the observed value. Per-axis sweeps and full per-country tables for each row appear in the subsections below and in Appendix A17.

Table 17: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
Dataset preprocessing irrelevant	JSD range over 6 configs	< 0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	< 0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	< 0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	< 0.010	**0.0077**
PT-IS $>$ best deterministic shift	ΔJSD(CS-Clamp$\to$PT–IS)	> 0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$< 8/15$	**3/15**
PT–IS kernel $\equiv$ ARGS-PT	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

A10.2 Default Hyperparameters and Validation

Table 18: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

Parameter	Symbol	Value
Persona agents	N	4
IS samples / pass	K_{half}	64 (total $2K_{\text{half}} = 128$)
Reliability scale	s	0.04
Perturbation std. dev.	σ	0.3
Cooperation weight	λ_{coop}	0.7
IS softmax temp.	η	0.5
ESS guard threshold	ρ_{eff}	0.1
PT curvature	$\alpha{=}\beta$	0.88
PT loss aversion	κ	2.25
Decision temp.	T_{dec}	0.5
T_{cat} (Species / Gender / other)	-	4.0 / 3.5 / 1.5
Default logit divisor	T_{logit}	3.0
ESS anchor blend	-	on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only)

Implementation. The dual-pass controller follows the equations in §3. Optional environment overrides can change K_{half}, reliability scale s, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters (σ, λ_{coop}, η, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix A10, Appendix A17) to show conclusions are stable beyond one tuned point.

A10.3 Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding T_{cat}[Species]$=4.0$, T_{cat}[Gender]$=3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the DISCA default (†) is strictly *more conservative* than the empirically best value ($\Delta\text{JSD} \in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 19: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	$+0.0013$
	0.25	0.0491	0.658	$+0.0002$
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	-0.0013
	1.00	0.0466	0.641	-0.0023
	2.00 (best)	0.0428	0.627	-0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	$+0.0052$
	1.0	0.0492	0.673	$+0.0023$
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	-0.0008
	3.0	0.0454	0.707	-0.0015
	4.0	0.0451	0.701	-0.0018
	6.0 (best)	0.0448	0.684	-0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	$+0.0039$
	1.00	0.0512	0.589	$+0.0025$
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	-0.0012
	3.00	0.0458	0.693	-0.0029
	4.00 (best)	0.0450	0.706	-0.0037
	JSD span		**0.0077**	

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near T_{dec}=1.0 and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep T_{dec}=0.5 as a deliberately conservative operating point.

A10.4 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters: s (reliability gate scale), λ_{coop} (individual-vs-consensus weight), σ (IS proposal floor), and T_{cat} (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold the other three axes at their defaults; the s axis directly probes how the choice of reliability scale in Eq. 6 affects downstream MIS.

Table 20: Hyperparameter sensitivity ranges on the three-country panel (Phi-4, n=250/country). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 3 are not the product of a particular hyperparameter choice.

Axis	Default	Grid	Min MIS	Max MIS	Δ
s	0.04	$\{0.01, 0.02, 0.04, 0.08, 0.16\}$	0.4091	0.4164	0.0073
λ_{coop}	0.70	$\{0.30, 0.50, 0.70, 0.85, 0.95\}$	0.4085	0.4173	0.0088
σ	0.30	$\{0.10, 0.20, 0.30, 0.45, 0.60\}$	0.4084	0.4122	0.0037
T_{cat} scale	1.0×	$\{0.33\times, 0.67\times, 1.0\times, 1.33\times, 1.67\times\}$	0.4060	0.4149	0.0090

Why these four axes. s controls the steepness of the reliability gate (Eq. 6); λ_{coop} trades off individual-persona PT utility against the consensus utility; σ floors the IS proposal scale so the N=4 empirical std cannot collapse to zero; and T_{cat} scales the whole family of per-category decision temperatures (preserving relative balance)-a global sharpening / flattening knob on the decision logits. Per-category temperatures are stress-tested separately in the temperature sensitivity sweep above.

Persona count, IS budget, and per-category vs. global T_{cat}. We additionally swept three operational knobs on the same USA/VNM/DEU panel. **Persona count** ($N \in \{2, 3, 4, 5, 6\}$, Table 21): N=4 is optimal at macro MIS 0.4051, with degradation at both smaller and larger panels (0.4252 at N=2, 0.4179 at N=6), supporting the four-persona design. **IS budget** ($K_{\text{half}} \in \{8, 16, 32, 64, 128, 192\}$, Table 22): performance is flat across an order of magnitude (best 0.4092 at K=16; 0.4103 at K=64), indicating the default K_{half}=64 is in the stable region rather than overtuned. **Global vs. per-category T_{cat}** (Table 23): the per-category default outperforms *every* global schedule we tested (0.4098 vs. all seven global values in $\{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$, which land in 0.4119–0.4138). The advantage is small in absolute terms but consistent.

Table 21: Persona-count sweep on the USA/VNM/DEU panel (Phi-4, n=250/country). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

N	Macro MIS ↓	Std (across countries)	Flip rate	sec/scenario
2	0.4252	0.1045	0.275	0.089
3	0.4194	0.0945	0.273	0.100
4	**0.4051**	0.0829	0.276	0.111
5	0.4107	0.0745	0.276	0.120
6	0.4179	0.0919	0.245	0.130

Table 22: IS-budget sweep (K_{half} samples per pass; total per scenario is K_{total}=2K_{half}). Macro MIS is flat across an order of magnitude. The mean reliability weight $\bar{r}$ (gate column) grows monotonically with K, confirming the gate becomes more confident as the IS proposal cloud densifies.

K_{half}	K_{total}	Macro MIS ↓	Flip rate	$\bar{r}$ (gate)	sec/scen
8	16	0.4114	0.284	0.684	0.121
16	32	0.4092	0.271	0.736	0.114
32	64	0.4104	0.278	0.766	0.114
64	128	0.4103	0.280	0.793	0.113
128	256	0.4147	0.274	0.814	0.116
192	384	0.4126	0.273	0.827	0.112

Table 23: Global vs. per-category T_{cat} on the USA/VNM/DEU panel. The per-category default (decision 4.0, social 3.5, non-social 1.5) outperforms every global value we tested. The gap is small (≤ 0.004) but uniformly directional, supporting category-specific decision temperatures over a single shared scalar.

T_{cat} schedule	Macro MIS ↓	Std	Flip rate
per-category default (4.0 / 3.5 / 1.5)	**0.4098**	0.0881	0.278
global $T_{\text{cat}} = 1.00$	0.4120	0.0921	0.292
global $T_{\text{cat}} = 1.50$	0.4119	0.0838	0.304
global $T_{\text{cat}} = 2.00$	0.4138	0.0778	0.301
global $T_{\text{cat}} = 2.50$	0.4134	0.0774	0.296
global $T_{\text{cat}} = 3.00$	0.4127	0.0749	0.303
global $T_{\text{cat}} = 3.50$	0.4138	0.0689	0.306
global $T_{\text{cat}} = 4.00$	0.4128	0.0734	0.302

A11 Scenario-Slice Release and Category-Cap Sensitivity

Section 4.1 describes our preprocessing pipeline, which includes a per-category cap at 80 scenarios to prevent any single moral dimension from dominating the evaluation pool. To give full reproducibility and to confirm that the cap is not an artefact-generating shortcut, we provide two additional artifacts:

- **Exact scenario-slice release.** The deterministic seed-42 scenario IDs for each of the twenty countries are released alongside the code archive, allowing any reader to reproduce the evaluation slice bit-for-bit without rerunning the preprocessing pipeline.
- **No-cap sensitivity run.** We re-run Phi-4 DISCA on the twenty-country grid with the per-category cap disabled. This run executes 440 scenarios per country (vs. the capped slice) and yields macro MIS 0.4051, macro JSD 0.0662, and mean Pearson r 0.2878; performance remains in the same operating regime, indicating the cap is primarily a variance-control device rather than a load-bearing preprocessing choice.

Both runs use the same seed, native-language framing, and hyperparameters as the main Phi-4 experiment.

A12 Theoretical Grounding

DISCA sits at the intersection of three theoretical traditions, each supplying one ingredient that the others lack. *Control as probabilistic inference* [Levine, 2018, Todorov, 2009] recasts optimal decision-making as posterior inference over an intention distribution; path-integral steering [Williams et al., 2018] instantiates that view with softmax-weighted perturbations that provably approximate the KL-regularised optimal control under mild regularity conditions [Todorov, 2009]. This is what licences us to read our Eq. 5 as an inference step rather than an arbitrary scalar interpolation, and it motivates the Gaussian proposal around $\bar{\delta}$ as the entropy-maximal default in the absence of stronger priors. *Prospect Theory* [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] then supplies the utility shape: the key empirical regularity-that human decision makers weight a loss of magnitude x roughly $\kappa{\approx}2.25$ times more than a gain of the same magnitude-is exactly the asymmetry we want an LLM cultural controller to respect, because a correction that silences one demographic persona is worse than a correction that fails to help it. Recent evidence that prospect-theoretic biases emerge naturally in LLM decision-making under risk [Payne, 2025] strengthens the match: the asymmetric gain/loss structure we impose *mirrors* one the models themselves already exhibit, rather than introducing an alien objective. *Importance-sampling (IS) theory* [Elvira and Martino, 2021] finally supplies the variance-control machinery, and it is here that self-normalised IS is known to be biased at finite K; the dual-pass construction we adopt is a light-weight bootstrap-style variance probe [Efron, 1979] that uses the same compute budget to detect instability rather than silently accept it, paired with an effective-sample-size guard grounded in design-effect theory [Kish, 1965]. We treat the dual-pass gate as a variance heuristic rather than a formal unbiasedness certificate; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A14).

A13 Design Rationale: Why Each Component

This appendix collects the full justification for the three design choices previewed in §1 and §3: the four-persona WVS panel, the Prospect-Theory IS kernel, and the dual-pass reliability gate.

Why a panel, and why the World Values Survey? A single "prompt-as-country" encoder-e.g., *"Answer as a Japanese person"*-collapses a distribution to a point estimate and discards the very spread the correction stage needs. Cultural-value research has long argued against this framing: both Inglehart-Welzel's modernisation theory [Inglehart and Welzel, 2005] and Schwartz's refined theory of basic individual values [Schwartz et al., 2012] find within-country variance on any single value dimension comparable to between-country variance, with demographic cleavages-most prominently age cohort-driving the largest slice. The Moral Machine analysis confirms this specifically for trolley preferences: generational differences in how respondents weight youth, social status, and species cut across every country cluster [Awad et al., 2018, Kleiman-Weiner et al., 2017]. WVS Wave 7 [Haerpfer et al., 2022] is the largest public instrument that reports *individual-level* microdata with the relevant age, gender, and value coverage, making it the natural substrate for data-driven personas rather than author-crafted stereotypes-a well-documented failure mode of prompt-based persona work [Deshpande et al., 2023].

Why $N{=}4$ specifically? Four is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance-young (<36), middle-aged (36–55), older (>55)-*plus* a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise. Three personas would sacrifice the population anchor and leave the consensus $\bar{\delta}$ vulnerable to a single outlier cohort; five or more would duplicate coverage of the same axis without adding a new source of variance (WVS does not resolve finer cohorts reliably for every country). The cohort structure matches the statistical-survey tradition of stratifying by age before any country-level estimate [Kish, 1965]. Empirically, the ablation in §6 confirms the ensemble is load-bearing: removing it costs -0.183 Pearson r and $+0.083$ MIS.

Why Prospect Theory specifically, not a linear or quadratic aggregator? Three independent considerations pick out the Kahneman-Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] over simpler alternatives. *(1) Asymmetric harm.* In a cultural panel, a correction that improves three personas while making one worse is not equivalent to one that helps all four equally: the former silences a demographic group the survey says exists. Linear averaging is oblivious to this asymmetry; quadratic aggregation treats $+x$ and $-x$ as equally valuable and costly. Loss aversion with $\kappa{\approx}2.25$ is the textbook object for encoding "harm is worse than equal-sized help". *(2) Diminishing sensitivity matches the scale of the cultural signal.* Gain curvature $\alpha{=}0.88{<}1$ compresses large persona gains relative to small ones, preventing one outlier persona (perhaps from sparse WVS coverage) from dominating the utility [Tversky and Kahneman, 1992]. *(3) Empirical compatibility with the base model.* Recent work shows that LLMs exhibit prospect-theoretic biases in decision-making under risk, with loss-aversion coefficients in the same neighbourhood as human respondents [Payne, 2025]. While we do not claim that logit-gap changes in an LLM are psychologically equivalent to monetary gains and losses in human subjects—the PT value function is used here as a *controller aggregation kernel*, not as a cognitive model of the LLM—the empirical compatibility means we impose a utility shape that does not fight the model's own implicit asymmetries. Auxiliary runs with linear reward kernels and WVS-weighted linear kernels on the same pipeline confirm that the PT shape outperforms symmetric alternatives, and the robustness sweeps in Appendix A10 show that the method's conclusions are stable across three logit-shift kernels, so the specific $\kappa{=}2.25$ is a well-motivated default rather than a sensitive tuning knob.

Why importance sampling at all? The cultural correction we seek is the argmax of a utility that values each persona's alignment and penalises corrections that make any of them worse [Levine, 2018]. Closed-form optimisation is unavailable because the utility is non-convex (PT's kink at zero, the ℓ_1-style gain definition in Eq. 13), so we approximate the optimal control distribution with a path-integral-style softmax over perturbations [Williams et al., 2018, Todorov, 2009]. The Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ plays the role of an entropy-maximal default, and the softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ implement the KL-regularised Boltzmann posterior balancing utility against deviation from the consensus. This control-as-inference reading licenses Eq. 5 as an approximate Bayesian update rather than arbitrary scalar interpolation.

Why dual-pass, rather than more samples or a hard threshold? Self-normalised IS is a biased estimator at finite sample size [Elvira and Martino, 2021]: the bias shrinks as $O(1/K)$ but does not reveal itself to the caller, so "more samples" hides instability rather than flagging it. Two independent replicates of size K_{half} using the same total budget $K{=}2K_{\text{half}}$ trade a small increase in per-pass variance for a *direct, scenario-level probe* of that instability: the squared gap $(\delta_1^\star - \delta_2^\star)^2$ is a one-dimensional bootstrap-style variance estimator [Efron, 1979] whose small values license the correction and whose large values veto it automatically, without a hand-tuned threshold. The exponential map $r{=}\exp\big(-(\delta_1^\star - \delta_2^\star)^2/s\big)$ is preferred over a hard indicator because it preserves a single differentiable code path and its decay matches the shape of a Gaussian likelihood ratio under independent noise. An effective-sample-size guard [Kish, 1965] on each pass ($N_{\text{eff}}/K_{\text{half}}{<}\rho_{\text{eff}}$) zeros the offset entirely when the weights collapse, providing a second, coarser safety net below the smooth reliability gate. We emphasise that this construction is a variance heuristic, not a formal unbiasedness certificate; finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A14).

A14 PT–IS, Sampling, and Dual-Pass Reliability

A14.1 Gain definitions

For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, the per-persona and consensus gains are:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{13}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{14}$$

$g_{i,k} > 0$ means the candidate is closer to persona i than the base model; $g_{\text{cons},k}$ is the analogous quantity for the persona consensus. These enter Eq. 5 after σ-normalisation. The IS update for each pass is $\delta_m^\star = \sum_k w_k^{(m)} \epsilon_k$ with $w_k^{(m)} = \mathrm{softmax}(U_{\text{total}}(\epsilon_k)/\eta)$ and temperature $\eta = 0.5$. An ESS guard zeros $\delta_m^\star$ when the Kish effective sample size $N_{\text{eff}}/K_{\text{half}}$ falls below $\rho_{\text{eff}} = 0.1$.

A14.2 Utility shape and the omitted explicit KL-on-ϵ term

The scalar decision gap δ acts as a one-dimensional state and the perturbations ϵ_k as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to U_{total} in Eq. 5: the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

A14.3 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 5 for clarity.

A14.4 Why split K into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-K bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

A14.5 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT → RIGHT immediately followed by RIGHT → LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A ↔ Group B in the native script. Logit gaps $(\delta_{\text{base}}, \delta_i)$ from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the released controller (debias first, then PT–IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final p_{spare} applies $\sigma(\cdot)$ only after that correction.

A15 Persona Construction Details

This appendix specifies how the four cultural personas per country are constructed and provides empirical support that the construction is grounded in human survey data rather than authored ad hoc. §A15.1 reproduces two personas verbatim from `generate_wvs_persona()`; §A15.2 documents the WVS Wave 7 processing pipeline; §A15.3 links the ten WVS cultural dimensions to the six MultiTP moral attributes both thematically and empirically; and §A15.4 shows that DISCA is robust to the choice of the fourth (anchor) persona.

A15.1 Example Personas

Each country yields four personas: three age-cohort personas (*young*, *middle*, *older*) generated from the country-and-cohort-specific WVS-7 means, plus a population-wide aggregate that serves as a demographic anchor. We illustrate the output with one English persona (United States, young-adult cohort) and one Vietnamese persona (Vietnam, older-adult cohort). The Vietnamese block is reproduced verbatim from `generate_wvs_persona()` and is followed by a sentence-by-sentence English gloss for non-Vietnamese readers.

United States, young-adult cohort (English, verbatim).

> *You are a young adult from the United States, in your 20s and early 30s. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are morally conservative on contested issues. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.*

Vietnam, older-adult cohort (Vietnamese, verbatim).

> *Bạn là một người cao tuổi đến từ Việt Nam, trên 60 tuổi. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn có niềm tin tôn giáo ở mức trung bình. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thỉnh thoảng tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ*

nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn rất thiếu khoan dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thế giới quan đã nêu ở trên.

Sentence-by-sentence English gloss.

You are a senior citizen from Vietnam, over 60 years old. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are leaning toward obedience and religious faith. On contested moral choices you are morally conservative on contested issues. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an occasional political participant. You are intensely proud of your country. Overall you are very happy with your life. On the role of women in society you are somewhat traditional on gender roles. In what you prioritise in life you are leaning post-materialist. Toward people unlike yourself you are strongly intolerant of outgroups. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.

Aggregate (fourth) persona. The fourth agent is constructed from the country's population-wide WVS profile, pooling respondents across age cohorts. It serves as a demographic anchor: by suppressing cohort-specific noise it ensures that the ensemble's consensus target $\bar{\delta}$ reflects the broadest available empirical signal rather than a single generational viewpoint. §A15.4 contrasts this choice with a country-invariant utilitarian anchor and shows that DISCA's macro-MIS is insensitive to the substitution.

Faithfulness disclaimer. The descriptors above are read directly from the country's WVS-7 means through deterministic quartile cuts (Table 24); they are not hand-selected to match cultural stereotypes. As a consequence, individual cohorts can present empirically grounded but locally counter-intuitive combinations (for example, a U.S. young-adult that is *leaning materialist* and *morally conservative on contested issues*), reflecting actual WVS-7 distributions rather than authorial bias.

A15.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the ten-variable cultural-value scheme of Greco et al. [2026]. The pipeline is fully deterministic: given the inverted WVS-7 CSV and the country code, it returns the four personas used at inference time without any tunable hyperparameters.

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 24). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* ($<$36), *middle* (36–55), *older* ($>$55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, each dimension's score is the unweighted mean of all valid (respondent, item) values associated with that dimension — i.e., for multi-item dimensions (e.g., Q177–Q182 for moral acceptability) every respondent contributes one value per item to a single pooled mean. For single-item dimensions this reduces to the standard across-respondent mean.
4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where (lo, hi) is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: ≥ 0.75 (strong positive), ≥ 0.50, ≥ 0.25, < 0.25 (strong negative). See Table 24.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 24: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: ≥ 0.75 (descriptor 1, strongest positive pole), ≥ 0.50 (2), ≥ 0.25 (3), < 0.25 (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

Dimension	WVS items	Range	Dir.	Descriptors ($\geq$0.75 / $\geq$0.50 / $\geq$0.25 / $<$0.25)
Religiosity	Q6P	1–4	$+$	deeply / moderately religious / somewhat / highly secular
Child-rearing	Q17P	0–1	$-$	independence-imagination $\leftrightarrow$ obedience-faith
Moral accept.	Q177–Q182	1–10	$+$	very permissive $\leftrightarrow$ strictly opposed
Social trust	Q57P	1–2	$+$	very high trust $\leftrightarrow$ deep distrust
Polit. particip.	Q199P, Q200P	1–3	$+$	active participant $\leftrightarrow$ non-participant
National pride	Q254P	1–4	$+$	very proud $\leftrightarrow$ not proud
Happiness	Q46P	1–4	$+$	very happy $\leftrightarrow$ not happy
Gender equality	Q29P, Q30P, Q31P, Q33P	1–4	$-$	strongly egalitarian $\leftrightarrow$ traditional roles
Materialism	Q152, Q153	1–3	$+$	post-materialist $\leftrightarrow$ materialist
Tolerance	Q19P–Q23P	0–1	$-$	very tolerant $\leftrightarrow$ intolerant

A15.3 WVS-to-Trolley Dimension Linkage

The linkage between the ten WVS cultural dimensions and the six MultiTP moral dimensions operates through two complementary mechanisms.

Direct thematic overlap. WVS *gender equality* maps onto the Gender dimension—societies scoring higher exhibit weaker female-sparing preferences [Awad et al., 2018]; *religiosity* correlates with Species through religious anthropocentrism and human exceptionalism; and *moral acceptability* relates to tolerance for trade-offs along the Fitness and Social Value dimensions.

Indirect modulation. Dimensions such as *social trust*, *child-rearing values*, and *materialism orientation* reshape the moral *frame* the persona adopts: post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values up-weight hierarchical social roles (affecting Age and Social Value). Importantly, DISCA does not require an exact causal mapping between WVS and MultiTP dimensions; the ten-dimensional persona ensemble produces diversity in logit-space gaps, and the importance-sampling stage selects the correction that balances collective utility.

Empirical validation of the WVS$\rightarrow$AMCE pathway. Table 25 regresses each of the six MultiTP human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standardised OLS). The 10-feature WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, with $|\beta| > 0.3$ in 31 of 60 cells. This is direct evidence that WVS-grounded persona construction is not arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA must reproduce.

Table 25: WVS features as predictors of human AMCE dimensions. OLS with standardised predictors. Coefficients are standardised (β); bold = $|\beta| > 0.3$. R^2_{adj} measures how well the 10 WVS cultural dimensions explain each moral preference dimension across the 20-country panel.

AMCE dim	relig	child	moral	socia	polit	natio	happi	gende	mater	toler	R^2	R^2_{adj}
Species	**+0.47**	+0.12	**-0.65**	**+0.60**	+0.08	+0.16	-0.02	**-0.74**	-0.30	**+0.40**	0.57	0.09
Gender	**+0.69**	**+0.37**	-0.18	**+0.41**	**-0.44**	**-0.39**	**+0.71**	**-0.81**	-0.20	+0.06	0.66	0.29
Age	**-0.80**	-0.09	**+0.61**	**-0.74**	+0.17	+0.26	-0.27	**+0.62**	**+0.49**	**-0.37**	0.55	0.05
Fitness	+0.28	-0.23	-0.16	-0.12	-0.17	**+0.43**	+0.09	**-1.46**	-0.21	**+0.53**	0.60	0.15
SocialValue	-0.27	-0.30	**+0.40**	**-0.40**	**-0.37**	**+0.58**	+0.17	**-0.31**	-0.12	-0.15	0.69	0.35
Utilitarianism	**+1.11**	**+0.39**	+0.07	**+0.59**	-0.21	-0.01	+0.13	**-0.76**	**+0.42**	+0.25	0.66	0.29

Table 26 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension d, we re-run DISCA with d removed from every persona and measure the change in per-MultiTP-dim absolute error (positive cell = error rises when the dimension is removed = that dimension is load-bearing for the corresponding moral attribute). The three largest positive couplings are *moral acceptability* $\rightarrow$ Social Value (+3.97 pp), *social trust* $\rightarrow$ Social Value (+2.81 pp), and *child-rearing values* $\rightarrow$ Utilitarianism (+2.41 pp). Cells with the top-3 largest positive couplings per row are bolded; rows with fewer than three positive cells are bolded only on those positive cells. Negative cells are also informative: *tolerance of diversity* $\rightarrow$ Social Value (-7.76) and *gender equality* $\rightarrow$

Gender (-3.42) indicate that, for those moral attributes, the corresponding WVS descriptor biases the persona ensemble *away* from the human target, so removing the descriptor improves accuracy. Crucially, DISCA does not require manual feature selection to handle these noisy dimensions: the loss-averse importance-sampling stage (Eq. 5) automatically down-weights candidate perturbations that worsen any persona's alignment, effectively acting as a built-in noise filter. When a WVS dimension introduces bias for a particular moral attribute, the PT-IS utility penalises the resulting correction asymmetrically (κ=$2.25\times$ for losses vs. gains), ensuring that noisy persona signals are suppressed rather than propagated. This self-correcting property is why the method remains robust despite imperfect WVS-to-trolley mappings—a design-level answer to the concern that the linkage is "indirect modulation" rather than causal.

Table 26: WVS-dim $\times$ MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS $\rightarrow$ MultiTP links).

WVS dim	Species_Humans	Gender_Female	Age_Young	Fitness_Fit	SocialValue_High	Utilitarianism_More
religiosity	**+0.66**	**+1.65**	-1.66	+0.58	**+1.02**	-0.37
child rearing	-1.82	**+1.29**	+0.86	**+1.30**	-0.33	**+2.41**
moral acceptability	**+1.13**	-0.02	-1.56	-0.34	**+3.97**	-1.69
social trust	+0.11	**+1.43**	-0.85	**+0.51**	**+2.81**	-1.88
political participation	-1.24	-0.82	-0.63	-0.36	-0.16	**+0.86**
national pride	-0.37	-0.10	-0.28	**+0.52**	**+1.19**	**+1.21**
happiness	-0.23	-0.88	**+0.13**	-1.69	-0.11	**+1.72**
gender equality	**+0.41**	-3.42	-2.90	-1.11	-1.87	**+0.62**
materialism orientation	**+1.35**	-1.26	**+0.90**	+0.04	-0.36	**+2.15**
tolerance diversity	-0.76	-3.52	**+1.95**	-0.59	-7.76	**+1.54**

A15.4 Sensitivity to the Fourth Persona

Setup. §3.2 pairs three age-cohort personas with a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that this choice is not load-bearing, we ran a head-to-head comparison of the two variants on the twenty-country Phi-4 grid.

Result. The utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 ($\Delta = -0.0203$), a marginal gain that does not change any qualitative conclusion of the paper.

Decision. We retain the aggregate variant as the default to preserve strict country-grounding of every persona and to avoid importing a globally fixed moral prior into country-conditioned steering. The utilitarian variant remains available as an opt-in for practitioners who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`).

A16 AMCE Estimation Details

Uniform treatment of all dimensions. All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism-are computed identically. For each dimension, we compute the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. **??**). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group ("More") or the smaller group ("Less") is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

Human AMCE scale. The MultiTP ground-truth AMCEs in the MultiTP-released country-specific AMCE table is on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

A17 Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country and stays below 5% on every country in the 20-country panel.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 27: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

A18 Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language]

\n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

A19 Baseline Implementation Details

A19.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for DISCA.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of DISCA is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

A19.2 ARGS and controlled decoding (relation to our work)

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 5 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 17 and §6 isolate the kernel comparison on the shared DISCA pipeline.

A20 Trigger Mechanism and Latency

DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the cross-backbone ablation suite (Table 5), indicating confidence modulation rather than wholesale decision reversal.

A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets: 0.086s/scenario at K=64, 0.083s at K=128, and 0.083s at K=256, compared with 0.023s for vanilla decoding (3.57–3.72× overhead).

Table 28: Inference latency benchmark (Phi-4, H100).

Method	K	sec/scenario	Overhead
swa_dpbr	64	0.086	3.72×
swa_dpbr	128	0.083	3.59×
swa_dpbr	256	0.083	3.57×
vanilla	–	0.023	1.00×

Cost-vs-quality frontier across backbones. Figure 5 plots the *deployed-cost* side of the Phi-4-vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-3.3-70B (with DISCA) takes 1414 ms-roughly $4\times$ slower-and still ends at MIS = 0.668. Phi-4 thus Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency. Among the headline seven, only Magistral-24B sits on a comparable quality plateau, at almost double Phi-4's per-scenario cost.

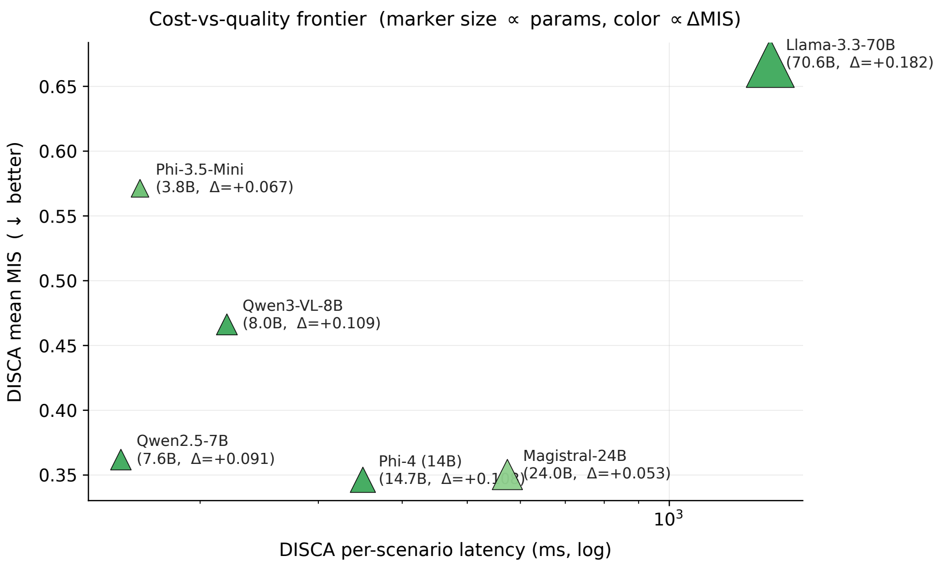

Figure 5: **Cost-vs-quality frontier on the headline 7 models.** Per- scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes ΔMIS (greener = larger DISCA gain). Phi-4 (14B, $\Delta = +0.108$) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

A21 Engineering Journey: Method Selection

DISCA (the "EXP-24" configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Table 29 summarises the top-performing variants.

Table 29: Method variant comparison on the 3-model $\times$ 5-country prototyping panel (15 rows). Mean MIS $\downarrow$. All variants share the same persona construction and debiasing; they differ in IS-stage design.

Rank	Method variant	Mean MIS $\downarrow$	Key innovation
1	EXP-23: Category-Coherence Reg.	0.3956	Blend IS toward per-cat mean
2	**EXP-24: Dual-Pass Bootstrap (DPBR)**	**0.3969**	Soft $r = \exp(-v_b/s)$ reliability
3	EXP-09: Hierarchical IS	0.3975	Cross-scenario smoothing (legacy)
4	EXP-22: Adaptive IS σ	0.3977	σ from agent/history stats
5	EXP-16: Nesterov IS	0.3977	Momentum + progressive PT
6	EXP-17: Dual-Momentum Prior	0.3979	Fast/slow β EMA blend
7	EXP-10: Grand Fusion	0.3982	EXP-03+05+09 combined
18	EXP-01: Base SWA-PTIS	0.4269	Single-pass IS baseline
25	EXP-12: Contrastive Personas	0.4517	World-average subtraction

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty signal** (r in Eq. 6) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows the value added by the dual-pass stabilization stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.

A22 Relationship to Persona-Dependent LLM Alignment

? is the closest prior work to DISCA in studying persona effects on LLM moral decisions. Their study and ours share the Moral Machine framework and AMCE-based evaluation, but differ fundamentally in goal, method, and scope. Table 30 summarises the key distinctions.

Table 30: Comparison between **?** and DISCA.

Dimension	Kim et al. (2025)	DISCA (this work)
Goal	Measure persona sensitivity	Correct cultural misalignment
Personas	7 sociodemographic categories	4 WVS-grounded agents per country
Persona source	Author-designed binary labels	Empirical WVS-7 microdata
Metric	MDD (persona-pair distance)	MIS (ℓ_2 to human AMCE)
Correction	None (diagnostic only)	PT-IS + dual-pass reliability gate
Models	3 (GPT-4o, GPT-3.5, Llama-2)	7 checkpoints, 5 families, 2B–70B
Countries	Aggregate (Western vs. Eastern)	20 individual countries
Scenarios	9 dimensions, 10k synthetic	6 dimensions, 310–500 per country
Theory	Partisan sorting (descriptive)	James–Stein shrinkage (Thm. 1)

Three findings from **?** directly inform DISCA's design.

Persona sensitivity motivates disagreement-driven steering. Kim et al. show that LLMs exhibit moral-decision distances (MDD) 2–4$\times$ larger than humans under contrasting personas. This confirms that persona prompts do shift LLM moral reasoning substantially, validating the premise that within-country persona spread is an informative signal. DISCA converts this sensitivity from a liability (uncontrolled decision shifts) into a feature (a sufficient statistic for correction reliability, Theorem 1).

The partisan sorting phenomenon justifies WVS grounding. Kim et al. find that political orientation produces the largest persona effect in LLMs, far exceeding its effect in human respondents. This "partisan sorting" effect reveals that LLMs over-index on politically salient dimensions when given sociodemographic labels. DISCA avoids this pathology by grounding personas in empirical WVS-7 microdata rather than author-designed binary labels: the ten-dimensional cultural profile (Table 24) distributes influence across religiosity, child-rearing values, social trust, and other axes, preventing any single dimension from dominating the correction. The leave-one-WVS-dim-out probe (Table 26) confirms that influence is distributed rather than concentrated.

Moral flips validate the reliability gate. Kim et al. document that specific personas can entirely reverse LLM preferences on certain moral dimensions (e.g., social status under a progressive persona in GPT-4o). DISCA's dual-pass reliability gate (Eq. 6) is designed precisely for this regime: when persona-driven corrections are large enough to flip the decision, the two independent IS passes are likely to disagree, and the exponential decay weight r shrinks the correction toward zero. The tail-safety analysis (Table 13) confirms this mechanism reduces harmed cells from 11/120 to 3/120 compared to ungated consensus.

From diagnosis to correction. The fundamental advance of DISCA over the Kim et al. framework is operational: where they diagnose persona sensitivity, we harness it. Their MDD metric measures the distance between two contrasting personas; our MIS measures the distance to the human target and actively reduces it. The James–Stein result (Theorem 1) provides the formal bridge: within-panel disagreement (analogous to their MDD) is a sufficient statistic for the correction's reliability, and the shrinkage estimator converts this diagnostic signal into an optimal correction.

A23 Extended Limitations

This appendix expands the limitations summarised in §6 with technical caveats not covered in the main text.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions operates through both direct thematic overlap and indirect modulation (Appendix A15.3). The 10-dimensional WVS profile explains $R^2 \in [0.55, 0.69]$ of human AMCE variance, and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones that the IS stage automatically down-weights. We avoid claims of the form "WVS dimension X caused AMCE shift Y," but the ensemble design and the IS noise-filtering mechanism together ensure the method is robust even when individual WVS–trolley links are imperfect.

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model-and when corrections diverge from targets-is open; we do not report that correlation matrix here.

Geographic coverage, preprocessing, and WVS vintage. We report 20 countries out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix A17. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice that may not reflect current preferences in rapidly evolving societies; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested, and one language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability-observed empirically on Unsloth Phi-4 prior to switching to vLLM BF16 for the headline runs.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 state three claims: (i) DISCA is a train-free inference-time controller that requires no per-country reward models or extra training data; (ii) it cuts macro MIS by 10–24% on seven open-weight backbones across 20 countries and 6 moral dimensions; (iii) calibration competes with scale (Phi-4 14B beats Llama-3.3-70B in absolute MIS). All three are substantiated in Table 3 and Figure 3; scope and assumptions (binary dilemmas with decision-token logits, WVS-7 vintage) are made explicit in §6 and Appendix A23.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §6 bounds generalisation with three caveats (alignment to a survey statistic vs. perceived legitimacy; PT value function as aggregation kernel rather than cognitive model; decision-token logit requirement). Appendix A23 expands these with: WVS-to-trolley linkage being indirect, the inter-persona reward assumption, geographic coverage (20 of 100+ MultiTP countries; sparse personas for under-sampled countries), WVS Wave 7 vintage, and 4-bit/Int8 quantisation artefacts on IS stability.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The single theoretical result, Theorem 1 (minimax-optimality of disagreement-driven shrinkage in the James–Stein sense for panels of $N \geq 4$), is stated in the main text with all assumptions (panel size, Gaussian persona shifts, ℓ_2 risk). The full proof, including referenced lemmas, is given in Appendix A1. A short proof sketch and the connection to Eq. 5 appear in §3.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.

- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: §3 gives the full algorithm (Algorithm 1, Eqs. 5–6); the full hyperparameter table (Table 18) lists every numerical default with the validation procedure (synthetic 200-scenario pool, transferability check on a 50-scenario MultiTP English subset). All backbones are publicly available open-weight checkpoints. Preprocessing (cap=80, oversample=36), seeds (3), and per-country preprocessing details are documented in Appendices A17, A7.1. We additionally release a self-contained reference implementation (`DISCA_Alignment/`) with the same defaults and a single-command sweep entry point.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: We release `DISCA_Alignment/` (anonymised for review; MIT-licensed) as supplemental material. The package is a self-contained slice of the research codebase:

a single CLI entry point (`run.py`) drives the per-model 20-country sweep, with backend selection (`vllm` / `hf_native` / `unsloth`) and explicit flags for the three external data paths (`-multitp-data`, `-wvs-data`, `-human-amce`). The README documents install (Python 3.10+, CUDA 12.x), HF token wiring for gated models, exact commands, and the output schema (per-country baseline / DPBR CSVs and a comparison file). MultiTP, WVS-7 Wave, and the human AMCE table are publicly available datasets cited in §4.1; access instructions are reproduced in the README.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: There is no training (the method is inference-time on frozen weights); we therefore detail all decoding and controller settings. §4 specifies the dataset (MultiTP, 20 countries), preprocessing (deduplication, category cap=80, oversample to $\geq$36 per dimension), the seven main backbones with sizes, and the metric definitions (MIS, Pearson r, JSD, win rate). All hyperparameters and how they were chosen are tabulated in Table 18 with the synthetic-pool grid-search procedure described in Appendix A10. Sensitivity sweeps for the temperature family (Appendix A10), category cap (Appendix A11), and the controller knobs (s, λ_{coop}, σ, T_{cat}) appear in Appendix A10.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The headline table (Table 3) reports macro MIS as mean $\pm$ standard deviation across 3 independent seeds (independent IS noise draws; the underlying model weights and data are fixed). A separate multi-seed stability analysis (Appendix A7.1) gives per-country variability, and a bootstrap confidence interval on JSD (± 0.004, Appendix A10) quantifies the noise floor for distributional metrics. We explicitly state that conclusions are stable well within the bootstrap noise floor; sensitivity sweeps along nine independent axes confirm this.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix A20 (Table 28) reports per-scenario wall-clock on a single H100 GPU: 0.083–0.086 s/scenario for DISCA at $K \in \{64, 128, 256\}$ vs. 0.023 s/scenario for vanilla decoding (3.57–3.72$\times$ overhead). Backbone-level cost is given in Figure 5: 350 ms/scenario for Phi-4 vs. 1414 ms for Llama-3.3-70B with DISCA. Sub-70B backbones run on a single H100 in BF16 via vLLM; the 70B backbone uses 4-bit Unsloth on the same hardware. The released code's CLI also runs end-to-end on a Kaggle T4/P100 dual-GPU node for sub-14B models. We acknowledge in Appendix A21 that exploratory experiments (25 controller variants on a 3-backbone $\times$ 5-country prototyping panel) consumed substantially more compute than the reported runs.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The authors have reviewed the NeurIPS Code of Ethics. The work uses only public, ethically-collected datasets (MultiTP [Jin et al., 2025], World Values Survey

Wave 7, Moral Machine human AMCEs [Awad et al., 2018]); it conducts no new human-subjects research and introduces no scraped data. Anonymity is preserved in the submission. Negative-impact considerations (risk of encoding harmful majorities) are explicitly discussed in §6 with a per-persona utility floor mitigation validated in Table 6.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §6 discusses both directions. Positive: a train-free, low-overhead controller lets practitioners respect cross-national preference heterogeneity at deployment time without finetuning per country, lowering the barrier to culturally pluralistic alignment. Negative: aligning to majority survey statistics could entrench harmful majorities or marginalise minority views [Zewail et al., 2026], and alignment to a survey statistic is not equivalent to legitimacy [Atari et al., 2023]. We propose and validate a per-persona utility floor that caps how far any single persona's post-correction utility can drop from vanilla (Table 6), and we are explicit that DISCA is a research artefact, not a deployment-ready normative system.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We do not release any pretrained models, generative models, or scraped datasets. The released artefact (`DISCA_Alignment/`) is an inference-time controller layered over publicly available open-weight LLMs whose existing safeguards remain in place.

The method itself produces no novel generative capability beyond that of the underlying models; the per-persona utility floor (Table 6) is included as an in-method safeguard against majoritarian skew.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All third-party assets are cited with version information. Datasets: MultiTP [Jin et al., 2025], the Moral Machine human AMCE table [Awad et al., 2018], and World Values Survey Wave 7 (`WVS_Cross-National_Wave_7_inverted_csv_v6_0.csv`) are used under the terms of the respective public releases. Models: Phi-4 and Phi-3.5-mini (MIT), Llama-3.3-70B (Llama-3 Community License), Qwen-2.5/3-VL (Apache-2.0/Qwen license), Gemma family (Gemma terms of use), Magistral-Small-2509 (Apache-2.0). Code dependencies (Transformers, vLLM, Unsloth, bitsandbytes) are listed with version pins in the released `requirements.txt`; we use them under their stated licenses (mostly Apache-2.0/MIT). The `compute_ACME` reference implementation we follow is credited to Jin et al. [2025] in Appendix A16.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: We release the DISCA reference implementation (`DISCA_Alignment/`) as a new code asset under the MIT License. The package ships with: a top-level README documenting installation, backend selection (vLLM / HF native / Unsloth), the three external data flags, environment variables, the quick-start command, and the output schema; a flat `disca/` module with one-purpose files (controller, dual-pass core, runner, AMCE/metrics, persona generation, multilingual scenarios, model loader); pinned dependencies in `requirements.txt`; and a single CLI entry point (`run.py -help`). The submission archive is anonymised. All hyperparameter defaults match Table 18 in the main paper.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper conducts no new crowdsourcing or human-subjects research. Human reference data is reused from the existing public Moral Machine experiment [Awad et al., 2018] as packaged by MultiTP [Jin et al., 2025], and from the World Values Survey Wave 7 public microdata; we make no new contact with participants and recruit no annotators.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No new human-subjects research is conducted. The work is a secondary analysis of pre-existing public datasets (Moral Machine human AMCEs, MultiTP scenarios, WVS-7 microdata), each of which carries its own ethics review at the original collection point; no further IRB approval applies.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.

- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: LLMs are central to the work in two ways, both fully described. (i) Frozen open-weight LLMs (Phi-4, Llama-3.3-70B, Magistral-24B, Qwen3-VL-8B, Qwen2.5-7B, Phi-3.5-mini, Gemma-4-E2B) are the *subject* that DISCA steers; their identities, sizes, and decoding settings are documented in §4.2 and Appendix A5. (ii) For the open-ended extension (§3.6), an LLM parses free-form responses into a (choice, confidence) pair to recover a pseudo-logit-gap; this auxiliary use is documented in that section. A separate synthetic 200-scenario tuning pool generated by GPT-4 is disclosed in Appendix A10 along with its limitations. LLMs were not used to generate the paper's claims, results, or proofs.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.